

Polling Place Location and the Costs of Voting ^{*}

Gaurav Bagwe[†] Juan Margitic [†] Allison Stashko[‡]

March 11, 2025

Abstract

We study how distance to one's polling place affects the likelihood of voting using a geographic regression discontinuity design with data from Pennsylvania and Georgia. A one-mile increase in distance to polling place reduces the likelihood of voting in person by 1 to 3 percentage points. Effects are two to three times higher among those closest to the polling place. When available, voters substitute to mail-in voting as distance to polling place increases. In counterfactual exercises, we identify turnout-maximizing polling places. Some precincts have large potential gains in turnout, even when choosing from existing buildings.

JEL Classification: D72, H70, K16

Keywords: Polling Places, Election Precincts, Costs of Voting

^{*}We would like to thank Laurent Bouton, Garance Genicot, Arik Levinson, Martin Ravallion, Carolina Concha Arriagada, J.J. Naddeo, Arturo Romero Yáñez, Daniel Valderrama, and Timothy Armstrong. We also thank participants at the American Political Science Association Annual Meeting, Southern Economic Association Conference, and the University of Utah Political Science Department. Thank you to Alvin Tsang and Sophia Bagley for their excellent research assistance.

[†]Bagwe and Margitic were doctoral candidates at Georgetown University while working on this project.

[‡]Emory University, e-mail: allison.stashko@emory.edu.

1 Introduction

Since the 2020 U.S. presidential elections, debates surrounding election administration have intensified. In 2021 alone, 36 states enacted a total of 96 new election laws.¹ Many of these laws shift the costs and benefits of voting in person or voting by mail. While some states made mail-in ballots available to all voters, others introduced new requirements for voting by mail. Some states expanded the number of early voting days, while others reduced the number of early voting days or removed polling places altogether. It is important to know how voters respond to changes in the costs of voting to assess the impact of these laws. This is especially true in the U.S., where voter participation is low and election law has become increasingly partisan in nature (Bentele and O'Brien 2013; Burden et al. 2017; Hasen 2012).

State and local government decisions regarding mail-in voting (Meredith and Endter 2016; Meredith and Malhotra 2011; Lockhart et al. 2020; Thompson et al. 2020), early voting, (Kaplan and Yuan 2020), and at-poll voting requirements (Highton 2017; Cantoni and Pons 2021) can each potentially have important consequences for voter participation. Even small changes to the convenience or cost of voting can determine whether or not someone votes, especially in large elections (Gomez et al. 2007; Braconnier et al. 2017).

This paper focuses on a particular cost of voting: the distance from a voter's home to their polling place. Distance to the polling place is an important determinant of voting behavior to study for three reasons. First, this cost of voting cannot be eliminated, unless we remove polling places altogether. The alternative to voting at a polling place is to vote by mail, though access to mail-in ballots varies from state to state. Nearly three-quarters of voters chose to vote at polls on election day in the 2018. During the COVID-19 pandemic, 54% of voters cast their ballot in person in the 2020 election cycle, and 67% did in 2024.² Second, the distance to polling place is inherently uneven among the population of eligible voters. It is important to understand if certain populations are systematically disadvantaged by higher costs of voting, in which case politicians may ignore their interests (Avery 2015; Martin 2003). Given the history of voter suppression in the United States, it is especially important to understand if the distance to polling place, and its effect on turnout, varies by race of eligible voters. Third, the polling place location is a policy choice, in contrast to other determinants of turnout like candidate characteristics or weather. State and local lawmakers determine how to divide a state into voting precincts and where to locate a

¹Brennan Center for Justice. Voting Laws Round-up: December 2021. Available at: <https://www.brennancenter.org/our-work/research-reports/voting-laws-roundup-december-2021>.

²Pew Research Center. Voters' and nonvoters' experiences with the 2024 election. December 4, 2024. Available at: <https://www.pewresearch.org/politics/2024/12/04/voters-and-nonvoters-experiences-with-the-2024-election/>.

polling places within precincts. A precise understanding of how polling place locations affect voter participation is critical for determining the optimal allocation of polling places.

Existing estimates of the effect of distance to polling place on voting vary widely.³ Evidence from panel data in North Carolina suggests that distance to polling place has a null effect on turnout (Clinton et al. 2020, Yoder 2018).⁴ Using matching methods to study polling place consolidations in Los Angeles County, Brady and McNulty (2011) find that a one-mile increase in distance to the polling place reduces turnout by 1 percentage point.⁵

At the other extreme, Cantoni (2020) estimates that a one-mile increase in distance to polling place reduces turnout by 4 to 12 percentage points in a study of nine municipalities in Massachusetts and Minnesota. These findings imply that moving a polling place one mile closer to a voter would cause an increase in turnout on par with some of the most successful turnout mobilization tactics (Gerber et al. 2017, Green et al. 2013, Enos and Fowler 2018). Cantoni (2020) identifies the causal effect of distance to polling place on turnout by exploiting variation in distance to polling place among voters near election precinct borders. Intuitively, registered voters who live close to an election precinct border are likely similar in terms of potential confounding factors but are assigned to different polling places. Tomkins et al. (2023) similarly use precinct borders as a source of variation. They estimate smaller effects: a mile increase in distance to polling place reduces turnout by 1.55 percentage points, on average, across 10 states.

The varied findings in the existing literature underscore the need for further study of polling places in the United States and their effect on turnout. One possibility is that the range of results reflects heterogeneous effects, given that each study uses data from different locations. Moreover, we know that there are large location and

³We focus on recent studies that use causal identification strategies. Polling places are typically located in schools and other government buildings that tend to be centrally located within a town. It is likely that eligible voters who live closer to polling places differ systematically from those who live further away in ways that matter for turnout. Earlier observational studies on the distance to polling place have found either a negative association with turnout or null results. Dyck and Gimpel (2005) find that distance to polling place is associated with lower turnout and more mail-in voting in Clark County, Nevada. Haspel and Gibbs Knotts (2005) find a negative relationship between distance to polling place and the likelihood of voting in Atlanta, Georgia. Amos et al. (2017) find a negative association between distance to polling place and voting in person on election day, but this is offset by a positive association between distance to polling place and voting early or absentee. Their study uses data from Manatee County, Florida.

⁴Alipour and Lindlacher (2023) find similar results by studying re-precincting in Munich.

⁵From Figure 3 of Brady and McNulty (2011), a change in polling place of one mile causes an estimated 4% reduction in the likelihood of voting. They estimate that most of this effect (-1.8%) is attributable to search costs of finding a new polling place, so the remaining -2.2% is attributable to distance to polling place. Given the reported turnout rate of 55.1% in their study, a 2.2% reduction in the likelihood of voting is equivalent to 1.2 percentage points.

election-specific effects on voting behavior (Cantoni and Pons 2022). However, there is also little overlap in methodologies used across existing studies, making comparisons difficult.

To help fill this gap, we collected information about the distance to polling place and turnout for over 15 million voters in Pennsylvania and Georgia. We chose two large swing states in order to estimate effects in urban and rural areas and among sociodemographic groups of interest. Regulations for accessing mail-in ballots also differ significantly between the two states, with Georgia providing easy access and Pennsylvania imposing more restrictions. For each registered voter in these two states, we observe the location of their residence, the location of their polling place, and whether or not they abstain, vote in person, or vote by mail in the 2018 election. We compute both the travel-route distance between a voter and their polling place and the time to travel to the polling place.⁶

To estimate the causal effect of distance to polling place on voting behavior, we use a geographic Regression Discontinuity (RD) design. Building on the innovation of Cantoni (2020), we use election precinct boundaries as a source of exogenous treatment assignment. We measure the distance from a voter to the nearest precinct boundary as the running variable. We observe 13,965 borders across the two states. The treatment variable, the distance from a voter to their polling place, introduces additional complexity compared to existing applications of geographic RDs, which typically have binary treatment variables (Keele and Titiunik 2015, Dell 2010). In particular, two voters in the same precinct can be equidistant from the border, but differ in distance to the polling place. To accommodate the continuous treatment variable, we use a fuzzy RD estimator. In the first stage, we estimate the change in distance to polling place at the discontinuity. In the second stage, we estimate the change in the likelihood that a registered voter votes.

A second challenge is that the monotonicity condition may not hold in this setting. Monotonicity requires that distance to polling place increases near the border when crossing from a control precinct to a treatment precinct (Card et al. 2015). Because distance to polling place naturally varies both across precincts and *within precincts* along the same boundary, the distance to polling place may increase near the border in one part of a precinct border, but decrease in another part.⁷ We resolve this issue by dividing borders into smaller segments, where necessary, such that the monotonicity

⁶Existing papers, in contrast, focus on Euclidean distance to measure the cost of travelling to the polling place. One concern is that Euclidean distance might differ systematically from travel-route distance in urban versus rural areas.

⁷The problem here stems from the fact that precincts are relatively small in size, so that polling places can be located near the discontinuity itself. This would not be an issue, for example, if using a fuzzy RD design to estimate the effect of the distance between units of observation and another location which is relatively far from the border.

condition is satisfied by construction. These methodological considerations are useful for future research that uses geographic RD to study the effects of a continuous treatment variable.

Our methodology has several advantages over existing approaches. The most closely related approach to ours is the main specification of Cantoni (2020). Cantoni estimates treatment effects by regressing voting outcomes on distance to polling place, including fixed effects for a small geographic area that contains a precinct border (a buffer zone of 0.05 miles around a segment of a border). We call this the Border Fixed Effects (FE) approach. Tomkins et al. (2023) similarly estimate a Border FE model, but they include fixed effects for neighborhood blocks.⁸ An important advantage of the RD approach is that treatment effects are estimated using only *across* precinct variation in distance to polling place. The Border FE approach uses both across- and within-precinct variation. Second, one can use data-driven methods to select voters within a bandwidth around a discontinuity, whereas the Border FE requires ad-hoc decisions about how to define a geographic fixed effect. Third, identifying assumptions for the RD estimator are perhaps more plausible. The RD requires continuity of confounding factors at the discontinuity, while the Border FE specification requires that distance to polling place is uncorrelated with confounding factors, after controlling for border fixed effects.

Relative to panel data methods, the RD approach allows us to isolate the effect of distance to polling place, separate from the effects of changes to polling places locations. Search costs (Alipour and Lindlacher 2023, Clinton et al. 2020, Yoder 2018, Brady and McNulty 2011) and habit formation in voting (Fujiwara et al. 2016; Plutzer 2002) make it difficult use changes in polling place locations over time to study the effects of distance alone on voting behavior.⁹ The RD approach, which relies only on cross-sectional data, is useful for policy-makers who are interested in the long-run effect of a polling place location on turnout, after short-run effects of a change in polling place dissipate.

We estimate that there is a small negative effect of distance to polling place on the likelihood of voting in person, on average, ranging from -2.8 to -1.4 p.p. per mile in Pennsylvania and Georgia, respectively. Importantly, the average effects mask substantial heterogeneity. In particular, voters close to the polling place are more sensitive to an increase in distance to polling place than voters that live further away.

⁸Tomkins et al. (2022) define a block as a group of voters that have similar addresses to each other, differing only in the last two digits of the building number. They only include blocks in which voters are assigned to two unique polling places. There are relatively few blocks split across a precinct border, so they pool across ten states. Estimates from pooled data across multiple states can be difficult to interpret, however, since states differ in rules about voting by mail.

⁹For example, a voter’s initial distance to polling place might still affect their voting decisions after a polling place changes, at least in the short run.

In Pennsylvania, a mile increase in distance to polling place causes a 7.7 p.p. decline in the likelihood of voting in person among voters in the bottom quartile of distance to polling place, who are 0.4 miles from the polling place, on average. In Georgia, voters in the bottom quartile, who are 0.9 miles from the polling place, are 2.8 p.p. less likely to vote in person per mile increase in distance to polling place. In contrast, we fail to reject null effects for voters in the top quartile of distance to polling place.

Pennsylvania and Georgia voters differ in their substitution to mail-in voting. In Pennsylvania, where voters were required to provide a justification for their absence, there is no significant substitution to mail-in voting as distance to polling place increases. In contrast, in Georgia, where obtaining absentee ballots requires no justification, there is a significant increase in the likelihood of voting by mail as the distance to the polling place increases. As a result, there is a statistically significant negative effect of distance to polling place on the overall likelihood of voting in Pennsylvania, and a moderately precise null effect in Georgia.

To reconcile our findings with previous studies, we use the methodologies from the literature, including the Border FE specification, a matching specification, and a difference-in-differences approach. By comparing results, we conclude that heterogeneous effects explain the discrepancies in estimated effect sizes across methodologies. Due to nonlinear effects, estimators that impose constant marginal effects of distance to polling place on the likelihood of voting will underestimate average effects. Location-specific factors also likely play a role. For example, we find larger effects in Philadelphia, on the same order of magnitude as those found in Boston and Minneapolis by Cantoni (2020).

We then turn to heterogeneous effects to better understand what factors are associated with greater sensitivity to polling place locations. The analysis of the two large states allows us to explore heterogeneous effects by demographic characteristics, party affiliation, economic variables, and transportation. We find that travel time and mode of transportation likely explain the heterogeneous effects, more so than sociodemographic factors.

The estimated effects of distance to polling place on voting behavior tell us how sensitive voters are to a change in the distance to their polling place. But how important are these effects in aggregate? How would turnout respond to realistic changes in polling places? To answer these questions, we study counterfactual allocations of polling places in Pennsylvania, the only state with an estimated negative effect of distance to polling place on overall turnout. We numerically solve a planner’s problem to maximize turnout in an existing voting precinct by choosing the location of a polling place, based on the locations of all voters as well as the estimated sensitivity of turnout to changes in distance to polling place. In a second exercise, we take existing infras-

tructure into account by compiling a list of candidate polling places (schools, libraries, churches, and police stations). We find the turnout-maximizing building and also consider what would happen if each precinct doubles the number of polling places, using existing buildings. We find that the polling places used in 2018 in Pennsylvania tend to be located near the turnout-maximizing polling place locations, with a median distance of 0.4 miles between them. Using the turnout-maximizing existing buildings, turnout would increase by 3 p.p. (6%) in aggregate. This is on par with the effects of two additional weeks of early voting (Kaplan and Yuan 2020). If each precinct were to double the number of polling places, using existing buildings, then turnout would increase by 9 p.p. (18%), which is similar to the effects of large scale political advertising campaigns (Enos and Fowler (2018)).

Our findings highlight some important lessons for studies of polling places in the future. First, the importance of polling place locations for voter participation will vary depending on the context. If the affected population largely owns cars and drives to work, then a one mile increase in distance to a polling place will likely have a small effect. On the other hand, a small change to a polling place in an area where people do not tend to drive can have significant effects on turnout. While polling closures in rural areas typically draw media attention,¹⁰ a change to polling location in an urban area could possibly have an even larger effect on turnout and should be evaluated carefully by election commissions.

Electoral design is also likely important. With two states, we cannot determine how state-level policies affect sensitivity of voting to polling place locations. However, evidence from Georgia suggests that the availability of no-excuse voting by mail is important for mitigating the cost of traveling to a polling place. A longer distance to the polling place makes a voter in Georgia less likely to vote at polls and more likely to vote by mail. In Pennsylvania, by contrast, fewer voters substitute to mail-in voting. Importantly, the response of voting in person to distance to polling place is similar in both states. Distance to polling place could deter voters from voting at polls, regardless of whether there is a convenient alternative method of voting.

2 Institutional Background

We study the 2018 primary and general elections in Pennsylvania and Georgia. We focus on two large swing states with publicly available voter registration and voter

¹⁰See for example, “Voting precincts closed across Georgia since election oversight lifted.” *Atlanta Journal-Constitution* August 31, 2018. Available at: <https://www.ajc.com/news/state--regional-govt--politics/voting-precincts-closed-across-georgia-since-election-oversight-lifted/bBkHxptlim0Gp9pKu7dfrN/>.

history data.¹¹ In 2018, there were 8.6 million registered voters in Pennsylvania and 6.9 million voters in Georgia.¹² In many states, including Pennsylvania and Georgia, 2018 was a year of historically high turnout for a midterm election (58% and 53% of registered voters cast ballots compared to 43% and 37% in 2014, respectively). Turnout was much lower for the primary elections (12% for Pennsylvania and 17% for Georgia).

The primary elections took place in May and the general elections on November 6, 2018. Both states elected their governor and all state executives, as well as members of the state legislature and the U.S. House of Representatives. One U.S. Senate seat was up for election in Pennsylvania and none were in Georgia.

Pennsylvania and Georgia differed in a number of election policies in 2018.¹³ In Pennsylvania, voting in person happened only on election day in 2018; there was no early voting. Voters also required an excuse to vote by absentee ballot at this time in Pennsylvania.¹⁴ In contrast, early voting began in Georgia three-weeks before election day and any voter could request a mail-in ballot up to 180 days before the election.¹⁵

Both states are divided into voting precincts by local government authorities (either county election commissions or municipal or county heads of government). An accessible location within each precinct, typically a school, library, police station, or church, is chosen as the polling place location by local authorities. Barring emergencies, polling places must be announced no less than 60 days prior to an election in Georgia and 20 days prior to an election in Pennsylvania. This means that voters might need to register to vote before knowing exactly where their polling place will be located. Neither state used same-day voter registration, so voters needed to register several weeks in advance of the election date. In Georgia, eligible voters were automatically registered to vote if they provided their information to the Department of Driver Services (e.g., to get a driver's license), unless they chose to opt out. In 2018, Pennsylvania had no such automatic registration system.

Importantly for our identification strategy, if voting in person, each voter could

¹¹The availability of voter registration files varies from state to state. See https://www.eac.gov/sites/default/files/voters/Available_Voter_File_Information.pdf for more information. By choosing large states, we have the ability to study heterogeneous effects across multiple dimensions. By choosing swing states, we study voters who are perhaps least sensitive to distance to polling place. If the benefit of voting is low in relatively uncompetitive elections, then voters may abstain regardless of distance to polling place. While we can not test this hypothesis directly, and questions of external validity remain, it is useful that Georgia and Pennsylvania had similar levels of turnout in the 2018 elections.

¹²Voter Registration Statistics from Pennsylvania's Department of State and Voter Registration Statistics from Georgia's Secretary of State Office.

¹³Election policies were retrieved from Pennsylvania and Georgia Secretary of State websites and from state election law: Pennsylvania Statutes Title 25 and Georgia Title Code 25.

¹⁴Pennsylvania introduced early voting and no-excuse mail-in voting in 2019, after the sample period, with Act 77.

¹⁵Georgia introduced no-excuse absentee voting in 2005 with House Bill 244. In 2021, the state added an ID requirement to vote by absentee ballot with Senate Bill 202.

only vote on election day at the polling place for the precinct in which they reside. Election day polls were open from 7am to 8pm in Pennsylvania and from 7am to 7pm in Georgia. If a registered voter had voted before in Pennsylvania, they did not need to bring identification. Georgia required voters to show photo identification when voting in person.

We do not have the ability to assess the effect of these electoral policies with only two states and one cross-section of the data. However, the policies are potentially important for understanding the substantive differences in our findings for Pennsylvania and Georgia.

3 Data

From the Pennsylvania Department of State and Georgia Secretary of State, we obtain voter registration files, which include a unique voter identification, address, and voting precinct for each registered voter within the state in 2018. We merge this information with the voter history files, which records whether or not a registered voter voted in each election as well as their method of voting (at polls or by absentee ballot).¹⁶

We find the locations of polling places from the Georgia Secretary of State’s website and from Pennsylvania’s state-run polling place look-up website.¹⁷ Next, we determine the latitude and longitude of polling place locations and registered voter addresses.¹⁸ We compute the travel-route distance, which is the shortest distance by road in miles between a voter and their assigned polling place.¹⁹ The travel route distance is preferred to Euclidean distance, since it takes road infrastructure into account. This is especially important since we have urban, suburban, and rural areas in our study, and road density might vary systematically with other factors that determine turnout. We also measure the Euclidean distance between voters and polling places and the travel time to the polling place, which is measured in minutes by car. To limit noise from imprecise geocoding, we exclude individuals that are more than 10 miles from their polling place (2.44 standard deviations from the mean in Pennsylvania and 3.09 standard deviations from the mean in Georgia). We also exclude individuals where there

¹⁶For Pennsylvania, we also observe whether the registered voter participated in previous elections. We also were able to observe voter locations in 2016 in Pennsylvania, but not in Georgia. We analyze the panel of voters in Pennsylvania in an alternative empirical specification in Section 4.2.

¹⁷Available at <https://www.pavoterservices.pa.gov/Pages/PollingPlaceInfo.aspx>. and <https://sos.ga.gov/index.php/elections>. Retrieved October 2018 and July 2020.

¹⁸To geolocate voters and polling places, we use the Address Locator provided by ArcGIS. This address locator uses interpolation to locate addresses, meaning it has the latitude and longitude of the endpoints of every street. It then interpolates the latitude and longitude of the specific address based on the street endpoints.

¹⁹Travel route distance and travel time were computed by the DIGIT Lab at the University of Utah’s Department of Geography, which uses ArcGIS.

is a discrepancy between the precinct to which they are geocoded and the precinct listed in voter registration files. All together, the observations that we exclude as likely geocoding errors account for 3% of registered voters in PA and 2% of registered voters in GA.

Finally, we assign voters to Census blocks using their geolocation, so that we can include additional information about the voter’s Census Block or Block-group from the 2010 Census and the American Community Survey (2006-2010 five year estimates), including income, education, and mode of travel to work. We also observe the voting age population of each block, which allows us to test whether distance to polling place affects the share of the voting-eligible population that is registered to vote (Section 5.2).

4 Identification Strategy

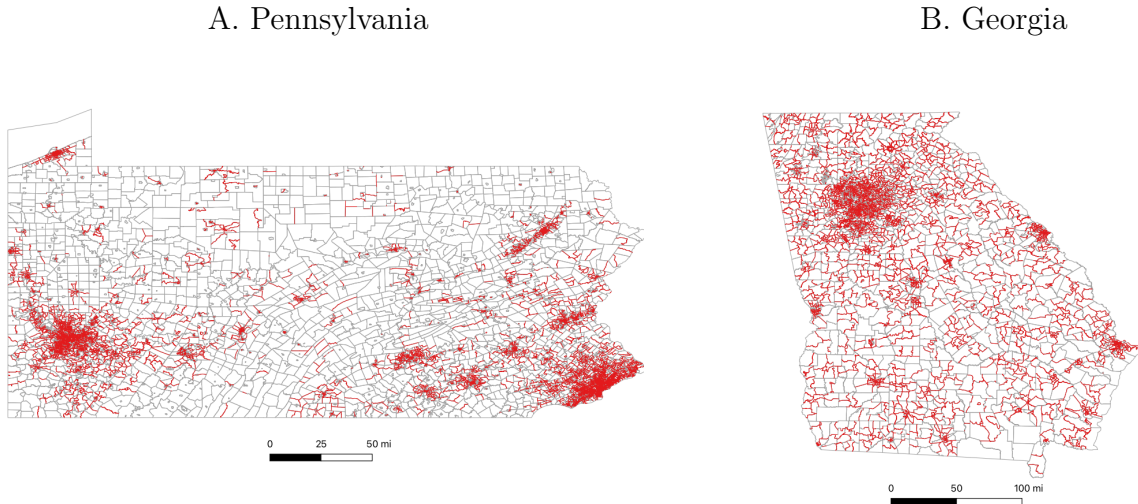
We estimate the effect of distance to polling place on the likelihood of voting in person, voting by absentee ballot (i.e., voting by mail), and voting by either method. Estimating this causal effect presents several challenges because a voter’s distance to their polling place is not random. Local election officials are supposed to choose convenient and accessible locations for polling places, often using schools and other public buildings. Voters who live close to polling places might differ systematically from voters who tend to live far away from polling places in ways that affect turnout decisions. For example, adults who choose to live close to a school tend to have school-age children and belong to a demographic group associated with a relatively low turnout rate (Wolfinger and Raymond 2008). These differences might be unobservable or not adequately measured due to the aggregated nature of some of the covariates in Census data.

To identify treatment effects, we use variation in distance to polling place created by election precinct borders. Election precincts are small geographic areas that assign voters to polling places. Intuitively, neighbors who live close to an election precinct border should be comparable in ways that affect voting behavior, but are assigned to different polling places and therefore may have different costs of voting in person. We focus only on a subset of precinct borders where this logic applies by discarding precinct borders that coincide with the borders of school districts, towns, counties, state legislative districts, or U.S. congressional districts.²⁰ Figure 1 shows the election precinct borders that are included and excluded for Georgia and Pennsylvania. There

²⁰We obtain 2010 voting tabulation district boundaries and all other political boundaries from the Census. There may be some differences between precinct boundaries reported to the Census in 2010 and implemented by the states in 2018. To account for this, we only include a border in the sample if voters on either side of the border are assigned to two different polling places locations in 2018.

are 9,432 borders included in the sample in Pennsylvania and 3,933 in Georgia.

Figure 1: Election precinct borders



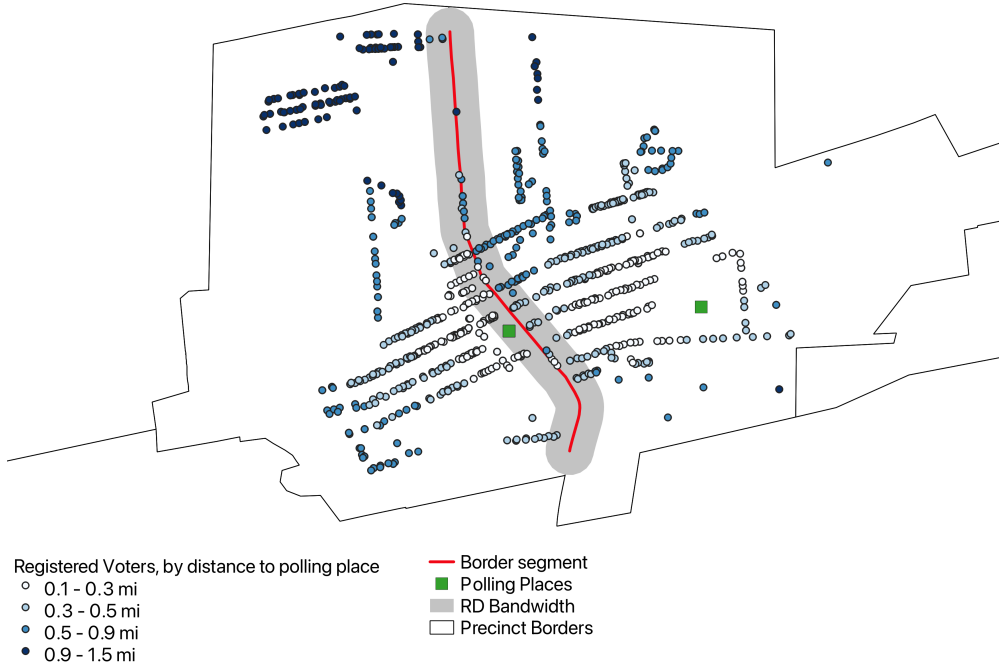
Note: These maps show election precinct borders in Pennsylvania and Georgia. Only precinct border segments highlighted in red are used in the empirical analysis. The red borders do not overlap with the borders of school districts, towns, counties, state legislative districts, or federal congressional districts.

We use two empirical strategies to estimate treatment effects using variation in distance to polling place near precinct border segments. The first, our preferred specification, is a geographic Regression Discontinuity (RD) design (Section 4.1). Treatment effects are estimated by comparing voters in close proximity to each other, but on opposite sides of a border segment. We complement the RD specification with a border fixed effects (FE) regression approach (Section 4.2) by following closely the methodology of Cantoni (2020). This allows us to compare our results to existing literature. We select a sample of voters close to a border segment, then regress voting outcomes on distance to polling place with border segment fixed effects.

The key distinction between the geographic RD and the border FE approach is that the geographic RD uses only *across*-precinct variation in distance to polling place, whereas the border FE approach uses both within and across precinct variation. Since polling places themselves can be located close to the border, especially in densely populated areas, there can be substantial variation in distance to polling place along one side of the border (see Figure 2). This highlights the challenges of choosing an appropriate geographic area to define fixed effects. For example, if the border is too long, then the within precinct variation in distance might be greater than the across precinct variation in distance. Moreover, it is not clear how to select an appropriate bandwidth around the border. Different bandwidths might be appropriate in densely

populated urban areas versus rural areas. Cantoni (2020) uses a buffer zone of 0.05 miles around a border, while Tomkins et al. (2022) use voters with similar addresses. If the fixed effects are not properly selected, then there may be bias from confounding factors. By comparison, in the RD approach, we only require continuity of confounding factors near the precinct border. Moreover, we can use existing methodologies to choose bandwidths and present graphic evidence in a data-driven manner. Finally, an advantage of the non-parametric RD estimation is that we impose no restrictions on treatment effect heterogeneity.

Figure 2: Example of a border segment used in the Geographic RD



Note: This figure shows the locations of voters and polling stations along one of the border segments included in both the regression discontinuity (RD) and border fixed effects (FE) specifications. This border is located in Union county, PA and is 1.2 miles long. Voters are shaded according to distance to the polling place. The running variable is distance to the border. All points on the border are closer to the polling place on the left than to the polling place on the right. This makes the left precinct the control precinct and the right precinct the treatment precinct. The RD sample includes all voters located within the 0.5 miles of the segment. The optimal bandwidth, indicated in gray, is 0.06 miles. The Border FE sample includes all voters located within 0.05 miles of the border segment.

4.1 Geographic Regression Discontinuity

In the RD specification, we use the exogenous jump in distance to polling place that occurs at the border of an election precinct to identify the effect of distance to the

polling place on voter turnout. Consider the following model of voting decisions:

$$Y_i = y(d_i, v_i, u_i), \quad (1)$$

Where Y_i is one of three outcome variables that indicate: whether an individual voted in person, whether an individual voted by mail, and whether an individual voted by either method. For ease of interpreting small coefficients, we have $Y_i=100$ if voter i votes and $Y_i = 0$ if voter i does not vote. Voting is a function of the distance between a voter and the polling place, d_i , a running variable, v_i , and an error term, u_i . This framework allows for a non-linear relationship between the outcome and distance to polling place and for treatment effect heterogeneity. We are interested in estimating the average marginal effect of distance to polling place on the likelihood of voting, $\partial y / \partial d_i$, called the local average response (Altonji and Matzkin, 2005) or local average treatment effect (LATE).

We construct the running variable, v_i , as the Euclidean distance between a voter i and the nearest point on a precinct border.²¹ We compute the running variable for all voters within 0.5 miles of any border.²² The running variable takes negative values for voters in the control precinct, where distance to polling place is smaller, and positive values for voters in the treatment precinct, where distance to polling place is larger. The challenge in this setting is that some borders have substantial variation in distance to polling place *within* each precinct. In such precincts, some voters are closer to their polling place near the border and others are further away, compared to voters on the other side of the precinct border. This makes it difficult to define treatment and control precincts. To resolve this issue, we cut borders into smaller pieces, called border segments, such that *all* voters sufficiently close to the border segment in the control precinct are closer to their polling place than *all* voters sufficiently close to the border segment in the treatment precinct. To do this, we draw a line connecting two polling places, A and B. The perpendicular bisector of this line divides the precincts into two areas, one in which voters are closer to polling place A and the other in which voters are closer to polling place B. Hence, we cut the border at any point where that the perpendicular bisector intersects the precinct border. All points along the resulting border segment will be closer to one polling place than the other. See Online Appendix A for details.

²¹As Keele and Titiunik (2015) point out, one might avoid using distance to the border as the running variable for long borders, e.g., the border of a nation or state. In that case, voters might be equidistant to the border yet far from each other. This could cast doubt on the continuity assumption. In our sample, the average length of a border is 1.1 miles in Pennsylvania and 2.6 miles in Georgia.

²²We could, in principal, compute the distance from every voter to every border. We avoid this due to computational challenges. The data-driven optimal bandwidths for the RD estimator are also substantially smaller than 0.5 miles.

With this running variable and definition of a border segment, we estimate effects of distance to polling place on voter behavior using a fuzzy RD approach. In the first stage, we estimate the change in distance to polling place at the border. In the second stage, we estimate the effect of distance to polling place on the likelihood of voting. The reason that we need a fuzzy RD instead of a sharp RD is that we can not write distance to the polling place as a function of the running variable alone. To see why, note that two voters can be equidistant from the border, yet differ in their distance to polling place.²³

As we have a fuzzy RD framework with a continuous treatment variable, we rely on the identification results of Card et al. (2015). The key identifying assumptions are that 1) the distributions of potential confounding factors are continuous at the border, 2) that the effect of the running variable on the outcome is continuous at the border, 3) the running variable is continuously distributed at the border, 4) there is non-zero first-stage effect, and 5) the monotonicity condition holds.

Monotonicity in this case requires that the distance to polling place is increasing in the running variable for all voters sufficiently close to the border. Typically, in a Fuzzy RD, the outside factors that affect treatment status are unobservable, so monotonicity is an assumption. Here, we observe all factors that determine treatment (the locations of the voter and polling place) and can therefore ensure that monotonicity is satisfied by the definition of both the running variable and a border segment. The necessary first-stage effect is also satisfied by construction.

We present evidence in support of continuity assumptions in Appendix A.2. Because we select only those precinct borders that have no administrative purpose except for on election day, we do not expect to see systematic differences in voter characteristics at the discontinuity. We test for discontinuities in the sex, age, and party affiliation of individuals. We also test for discontinuities in the census block or block-group level variables: median household income, whether it is an urban block, percent White, percent Black, percent Hispanic, percent with no high school diploma, percent that walk to work, and the number of cars per household. There are small but statistically significant discontinuities for age, sex, and cars per household in Pennsylvania and for likelihood of living in an urban block in Georgia. However, none of the RD coefficients are statistically significant after adjusting for multiple hypothesis testing. In Appendix A.2 we also investigate evidence of manipulation of the running variable.

Under the identifying assumptions, the fuzzy RD identifies the average of the treat-

²³An alternative to the fuzzy RD with a single-dimensional running variable would be a two-dimensional sharp RD. We choose to instead simplify a two-dimensional RD to a one-dimensional RD, in line with the recommendations of Imbens and Zajonc (2011) and Keele and Titiunik (2015). The simplification from a two-dimensional RD to a one-dimensional RD means that we can rely on existing identification results for continuous treatment variables, we can easily pool across many borders, and present visual evidence of the discontinuity.

ment on the treated for a single border segment, b . The fuzzy RD estimand is the ratio of the change in the likelihood of voting at the border over the change in distance to polling place at the border:

$$\tau_b^{RD} = \frac{\lim_{v \rightarrow 0^+} E[Y|v = 0] - \lim_{v \rightarrow 0^-} E[Y|v = 0]}{\lim_{v \rightarrow 0^+} d_i - \lim_{v \rightarrow 0^-} d_i}. \quad (2)$$

To estimate average treatment effects across borders, τ^{RD} , we follow recommendations of Imbens and Zajonc (2011) and Cattaneo, Titiunik, and Vazquez-Bare (2020) by pooling across border segments.²⁴ To improve precision we drop border segments with fewer than 50 observations and we partial out border segment fixed effects from all outcome variables. In the final RD sample, we have 9,674 border segments in Pennsylvania and 5,207 border segments in Georgia. We estimate RD coefficients using local linear regression with the MSE-optimal bandwidths of Calonico, Cattaneo and Titiunik (2015).²⁵

4.2 Fixed Effects Regressions

For the border fixed effects specification, we select all voters within 0.05 miles (264 feet or 80 meters) of any border segment.²⁶ We then estimate the following equation:

$$Y_i = \delta_{s(i)} + \beta d_i + X_i' \gamma + u_i, \quad (3)$$

where Y_i , d_i , and u_i are as before, $\delta_{s(i)}$ are border segment fixed effects, and X_i is a set of covariates for individual i .²⁷ The identifying assumption is that all unobservable factors affecting the likelihood of voting are uncorrelated with distance to polling place, within registered voters that are assigned to the same border segment.

²⁴Keele and Titiunik (2015) propose estimating τ_b^{RD} at several points along the border, using distance to the point as the running variable, and averaging across these points. This is feasible in our setting and produces similar results, though estimates are less precise when estimating the RD using observations near to an individual point, rather observations near to the full length of the border. In their application, they consider several points along a single long border. Here, we have many small borders with a relatively low sample size at each border.

²⁵We use the Stata command *rdrobust*. To ensure that results for the three voting outcomes are comparable, we use the same bandwidth for the likelihood of voting in person, by mail, and overall. We choose the MSE-optimal bandwidth for the likelihood of voting at polls. In Online Appendix B we show that RD estimates are not sensitive to the choice of bandwidth.

²⁶If a voter is within 0.05 miles of two segments (e.g., in a corner of a precinct), then they are assigned to the closer of the two segments. For each segment, we only include voters that are assigned to one of the two polling places for the precincts separated by the border segment (e.g., we exclude voters near an endpoint of the segment that are in a third precinct).

²⁷We include age, registered Democrat, and registered Republican. In addition to these individual-level covariates, we include census block- and block-group-level covariates: percent female, percent urban, percent White, non-Hispanic, percent Black, non-Hispanic, percent Hispanic, median household income, percent without high school diploma, cars per household, percent who commute to work by walking.

In a second more conservative fixed effects approach, we pair individual voters with the nearest voter on the other side of the border.²⁸ The estimating equation is the same as 3, but with fixed effects for each matched pair, rather than for a border segment. Thus, variation used to estimate treatment effects comes from only *across* precinct variation in distance to polling place, like with the RD approach. The identifying assumption is that, within each matched pair, distance to polling place is unrelated to confounding factors.

In Appendix A.3, we report balance tests for both fixed effects specifications. We regress individual- and census block-level covariates on distance to polling place, including either border fixed effects or matched pair fixed effects. Point estimates are similar to RD coefficients in Appendix A.2, though more precise. Given the large sample size, it is perhaps unsurprising that we find some statistically significant correlations between distance to polling place and observable characteristics.

5 Results

We first report the RD estimates for 2018 general elections in Pennsylvania and Georgia (Table 1). The RD plots in Figure 3 show corresponding graphical evidence for both the first-stage and reduced form results.

In Pennsylvania, the average increase in distance to the polling place at the discontinuity is 0.38 miles (SE=0.02). The estimated effect of distance to polling place on at-poll voting is -2.8 (SE=0.5), meaning that for every additional mile of distance to polling place the likelihood of voting in person falls by about 3 percentage points (p.p.), a 5% reduction. There is a small and statistically insignificant increase in voting by absentee ballot (0.3 p.p., SE=0.1), such that the estimated decrease in the overall likelihood of voting is 2.6 p.p. per mile (SE=0.5).

In Georgia, the average increase in distance to the polling place at the discontinuity is 1.04 miles (SE=0.03). The estimated effect of distance to polling place on at-poll voting is -1.4 p.p. (SE=0.3), a 5% reduction. There is a similarly sized increase in absentee voting (1.1 p.p., SE=0.3), such that the estimated effect on the overall likelihood of voting is small and statistically insignificant (-0.3 p.p., SE=0.4).

In both states, we see similar results if we measure the effects of an increase in Euclidean distance to polling place (Online Appendix C). We also estimate the effect of an increase in travel time to the polling place by car. Time to polling place at the discontinuity increases by 1 minute in Pennsylvania and by 2 minutes in Georgia. The

²⁸Each voter is paired with the nearest voter in the other precinct that is also within 0.05 miles of the border segment. Voters are matched with replacement. Since a single voter can belong to multiple matched pairs, we cluster standard errors by border and precinct, following Dube et al. (2010) and Cantoni (2020).

second-stage estimates imply that the likelihood of voting in person falls by 1.1 p.p. (SE=0.2) in Pennsylvania and by 0.8 p.p. (SE=0.2) in Georgia per minute of increase in travel time by car (Appendix C). We also find similar effects for the 2018 primary election in each state and from the 2016 general election (available for Pennsylvania only, Online Appendix D).

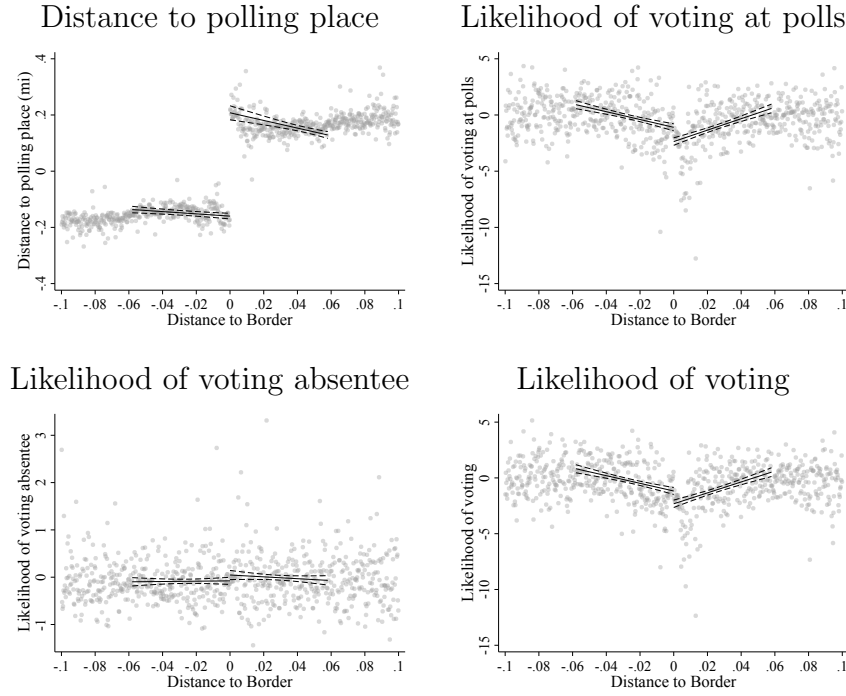
Table 1: Fuzzy RD Estimates

<i>A. Pennsylvania</i>				
	First-stage	Second-stage		
	Distance (mi)	At Poll	Absentee	Total
RD Estimate	0.38*** (0.02)	-2.84*** (0.53)	0.26* (0.14)	-2.58*** (0.53)
N	2,981,089	2,981,089	2,981,089	2,981,089
Effective N, Left	409,825	530,905	530,905	530,905
Effective N, Right	369,988	472,628	472,628	472,628
Bandwidth	0.04	0.06	0.06	0.06
Outcome mean	0.87	57.33	2.28	59.61
<i>B. Georgia</i>				
	First-stage	Second-stage		
	Distance (mi)	At Poll	Absentee	Total
RD Estimate	1.04*** (0.03)	-1.37*** (0.25)	1.12*** (0.34)	-0.25 (0.38)
N	2,968,151	2,968,151	2,968,151	2,968,151
Effective N, Left	463,005	380,239	380,239	380,239
Effective N, Right	415,559	342,894	342,894	342,894
Bandwidth	0.11	0.09	0.09	0.09
Outcome mean	1.90	25.74	28.85	54.59

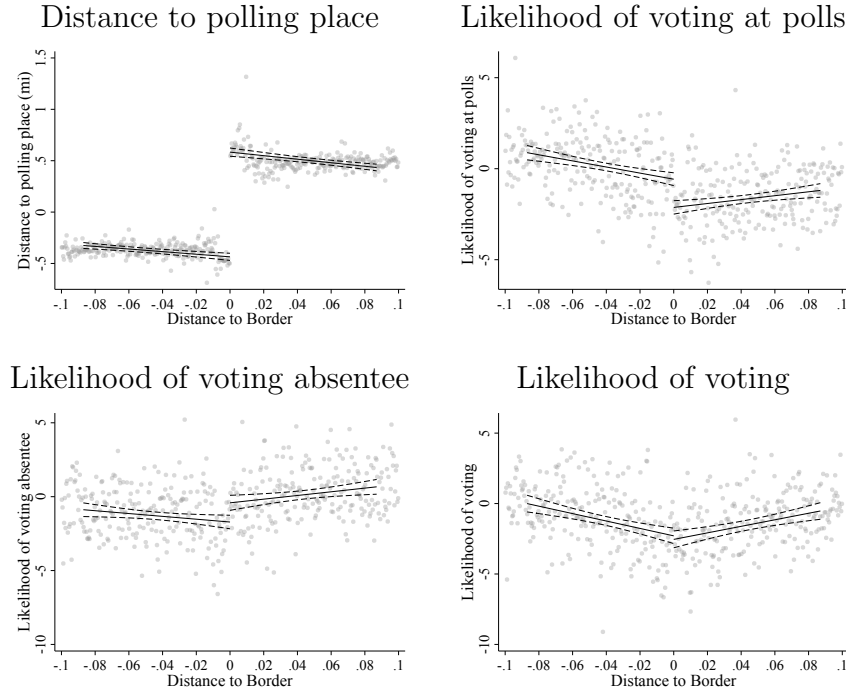
Note: This table reports RD Estimates for distance to polling place and voting outcomes. Voting outcomes indicate whether a registered voter voted in person (at poll), by mail (absentee), or by either method (total). Outcomes are scaled so that coefficients are measured in percentage points. Distance (mi) measures the travel route distance between a voter and their assigned polling place. Bandwidths are MSE-optimal for distance and at-poll voting. We use the same bandwidth for all voting outcomes. Standard errors allow for clustering at the border segment level.

Figure 3: Discontinuities in distance to polling place and likelihood of voting

A. Pennsylvania



B. Georgia



Note: This figure shows binned scatter plots and linear regression lines with 95% confidence intervals to the left and right of the discontinuity for the sample within the RD bandwidths. Distance to the border is the shortest Euclidean distance between a voter and the border segment. Distance to the polling place is the travel route distance between the voter and polling place (mi). The y-axes measure deviations from the border segment mean (either miles or percentage points). We use the Calonico, Cattaneo, and Titiunik (2015) mimicking variance evenly-spaced method to select bins.

The RD estimates suggest that an increase in distance to polling place deters registered voters from voting in-person. In Pennsylvania, this translates into a decline in overall turnout. In Georgia, voters substitute to mail-in voting so that there is a small and statistically insignificant effect on overall turnout. The 95% confidence interval rules out any decrease in turnout greater than 1 p.p.. It is important to remember that Pennsylvania and Georgia differed in their requirements for voting by mail in 2018. Pennsylvania required voters to provide an excuse for needing an absentee ballot, while Georgia did not. Without variation in absentee voting policies during the study period, we can not directly test the hypothesis that the availability of no-excuse absentee voting drives the higher uptake of absentee voting in Georgia.²⁹ However, the estimates are consistent with a larger share of voters substituting to mail-in voting if it's convenient to do so in response to a change in the cost of voting at polls.

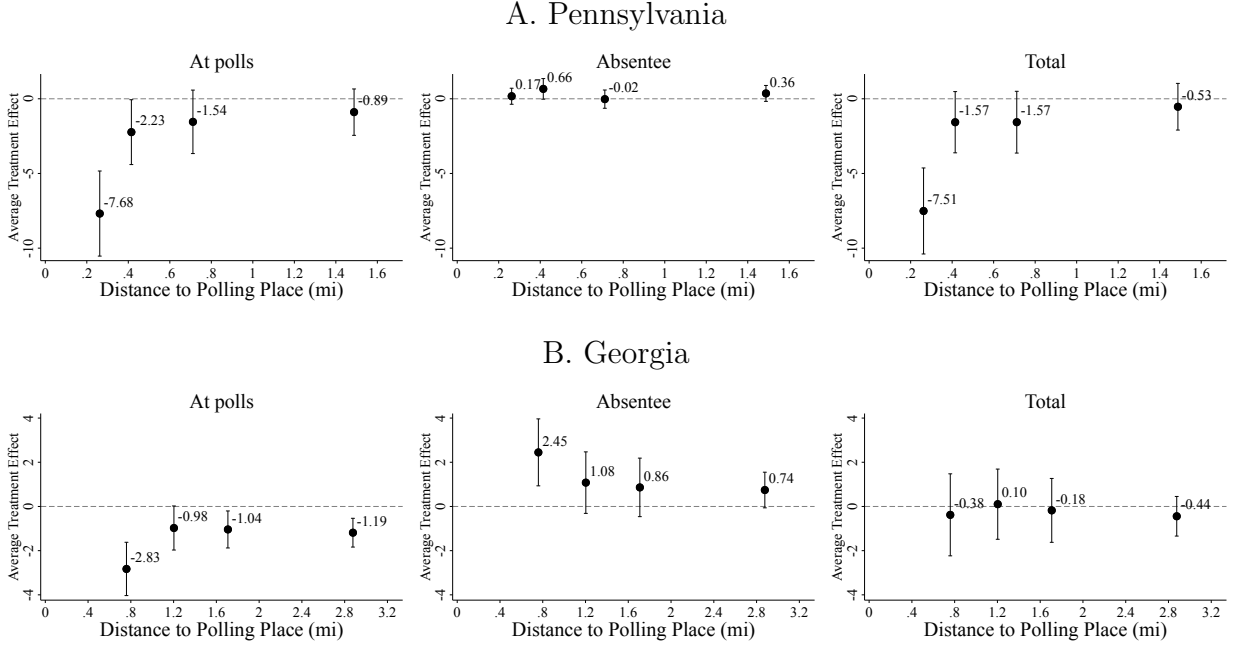
To interpret the magnitude of the estimated effects, and to compare estimates to existing studies in different contexts, we next consider the possibility of nonlinear effects. Although the non-parametric RD approach allows for a non-linear relationship between distance to polling place and voting decisions, such heterogeneity makes it difficult to interpret the LATE estimates.

5.1 Nonlinear effects

There are at least two reasons to expect non-linearities in this setting. It could be that the costs of voting in person are increasing in distance to polling place, as with a standard convex cost function. In that case, marginal effects would be larger in magnitude for voters further away from their polling place. Alternatively, voters who live close to a polling place may be more sensitive to distance to the polling place if they are less likely to use a car for transportation. Small changes in distance to polling place could be more costly for voters who walk or use public transportation than for those who use cars.

²⁹The switch from excuse-only to no-excuse absentee voting had no immediate effect on the sensitivity of voters to distance to polling place in the case of Minnesota (Cantoni 2020). In Georgia, no-excuse absentee voting may function differently, since it was a long-standing policy by the time of the 2018 election.

Figure 4: Nonlinear effects: Fuzzy RD estimates by distance to polling place



Note: This figure shows binned RD estimates. The RD sample is divided into quartiles based on the average distance to the polling place among observations in the control precinct of a border segment. The y-axis is measured in percentage points of likelihood of voting. The x-axis shows the average distance to polling place within each bin. Vertical lines indicate 95% confidence intervals. Standard errors allow for clustering at the border segment level.

To test for non-linear effects, we divide the RD sample into quartiles based on the average distance to the polling place among observations in the control precinct of a border segment (Figure 4 and Tables E.1 and E.2).³⁰

In both states, there is a pattern of non-linear effects in which voters closest to polling places are more sensitive to distance to the polling place than all other voters. In Pennsylvania, voters in the bottom quartile (who are 0.4 miles from the polling place on average) are 7.7 p.p. (SE=1.5) less likely to vote in person per additional mile of distance to polling place. In contrast, there is a statistically insignificant reduction of 0.9 p.p. (SE=0.8) likelihood of voting at polls per mile among for voters in

³⁰Note that each border segment identifies the average marginal effect $\partial y / \partial d_i$ at a level of d_i between the distance to the polling place in the control and treatment observations. We show that results are similar if we use the treatment observations to define bins instead in Online Appendix E. In all non-linear estimates, we pool across borders that vary in the first-stage effect. In the presence of non-linear effects, one concern is that border segments with very large first-stage effects might overestimate (with increasing marginal effects) or underestimate (with decreasing marginal effects) the treatment effect. A separate concern is that border segments with small first-stage effects could generate bias due to a weak identification problem. To address these concerns, we show that estimates in Figure 4 are not sensitive to excluding border segments with the largest and smallest first-stage effects (Online Appendix E).

the top quartile (who are 1.7 miles from the polling place on average).³¹ In Georgia, voters in the bottom quartile (who are 1.0 miles from the polling place on average) are 2.8 p.p. (SE=0.6) less likely to vote in person per additional mile to polling place, compared to 1.2 p.p. (SE=0.3) for voters in the top quartile (who are 3.3 miles from the polling place on average.)³² Similar to the results in Table 1, the effect of distance to polling place on voting by mail is small and statistically insignificant for each quartile in Pennsylvania, whereas there is positive, and fully offsetting, positive effect on voting by mail for each quartile in Georgia. One likely explanation for the nonlinear effects for in-person voting is that the mode of transportation differs for those who live relatively close to the polling place. We return to this hypothesis in Section 5.3.

Given the presence of non-linear effects, we use the estimates in Figure 4 to interpret the magnitude of the effects in each state, rather than the estimates in Table 1. The average of the point estimates in Figure 4 imply that, in Pennsylvania, a mile increase in distance to polling place causes a 3.1 p.p. reduction in the likelihood of voting at polls and a 2.8 p.p. reduction in the likelihood of voting overall. In Georgia, a mile increase in distance to polling place causes a 1.5 p.p. reduction in the likelihood of voting at polls and a 0.2 p.p. reduction in the likelihood of voting overall.

Another way to interpret the nonlinear point estimates is to consider the average change per standard deviation change in distance to polling place, across the four bins. This approach accounts for the fact that voters who are closest to their polling place tend to be in small precincts. Thus, voters who are the most sensitive are also less likely to experience large increases in distance to polling place.³³ Measured this way, the effects of distance to polling place on voting at polls are similar across states. A one standard deviation increase in distance to polling place decreases the likelihood of voting in person by 1.4 p.p. and 1.6 p.p. in Pennsylvania and Georgia, respectively. The corresponding change in the likelihood of voting overall is -1.2 p.p. in Pennsylvania and -0.3 p.p. in Georgia.

5.2 Fixed Effects Regression Results

In Table 2 we report estimated coefficients for specifications that include precinct fixed effects (Panel A), border segment fixed effects (Panel B), or matched pair fixed effects (Panel C). Coefficients are fairly stable across the three specifications in both Pennsylvania and Georgia. Focusing on the border fixed effects specification (Panel

³¹The estimates for the effect of distance on the likelihood of voting in person and the likelihood of voting overall in the bottom quartile are statistically significantly different from all other quartiles at the 1% level (Online Appendix E).

³²The estimate for the effect of distance on the likelihood of voting in person in the bottom quartile is statistically significantly different at the 5% level from estimates for all other quartiles (Online Appendix E).

³³We report standard deviation in distance to polling place by bin in Online Appendix E.

B), the estimated effect of distance on the likelihood of voting at polls is -0.6 (SE=0.2), with 95% confidence intervals ruling out a decrease greater than 1.1 percentage points in magnitude. There is an estimated 0.1 p.p. increase in the likelihood of voting by mail (SE=0.1) and a 0.6 p.p. decrease in the overall likelihood of voting (SE=0.2). In Georgia, the border FE estimates are -1.2 p.p. (SE=0.1) for voting in person, 1.4 p.p. (SE=0.1) for voting by mail, and 0.2 p.p. (SE=0.2) for the overall likelihood of voting. Hence, estimates from the fixed effects specifications are similar to RD estimates for Georgia, but substantially smaller in magnitude for Pennsylvania.

We argue that the presence of nonlinear effects can explain the differences between the estimated LATEs in Table 1 and ATEs in Table 2.³⁴ The LATEs estimated by the RD approach represent the weighted average of the marginal effect of distance to polling place on the likelihood of voting, with higher weights for observations closest to a border segment. The ATEs estimated by the FE specifications can also be interpreted as weighted average treatment effects, using the weighting results of Angrist and Pischke (2009). Observations have relatively higher weights if they belong to a border segment or matched pair with large variation in distance to polling place. Given the non-linear effects in Figure 4 one under-estimates the average marginal effect when up-weighting borders with large differences in distance to polling place. Indeed, even if we estimate nonlinear effects using sub-samples of voters by distance to polling place, we estimate smaller effects relative to the binned RD estimates (Online Appendix E).

Several robustness checks confirm that nonlinear effects, and other sources of treatment effect heterogeneity, can reconcile our findings with those in the existing literature. To start, heterogeneous effects can explain why estimates in Table 2 are an order of magnitude smaller than those in Cantoni (2020), even though the border FE methodology is essentially the same.³⁵ In Online Appendix G, we select Census Blocks in Pennsylvania and Georgia that are similar, based on observables, to the Census Blocks in the Minneapolis and Boston areas (using replication data from Cantoni, 2020). Point estimates are similar for the matched sample, which contains more urban areas, and the original sample of Cantoni (Table G.1). In general, voters tend to be more sensitive to distance to polling place in large cities with shorter distances to polling place (see Online Appendix H for estimates specific to Philadelphia, Pittsburgh, and Atlanta). Given the differences in the estimated effects of distance to polling place within a state, we further investigate heterogeneous effects next.

³⁴Confounding factors are unlikely to explain the discrepancies between RD and FE estimates, since point estimates change very little when we estimate effects using variation within a precinct (Panel A), within voters that are within 0.05 miles of a border segment (Panel B) or within a matched pair of voters (Panel C).

³⁵The only meaningful difference in methodologies for the FE specification is that Cantoni estimates turnout rates for parcels of land or Census Blocks, whereas we focus on individual likelihood of voting. We also estimate effects at the Census Block level in Appendix F.

Table 2: Fixed Effects Estimates

<i>A. Precinct Fixed Effects</i>						
	Pennsylvania			Georgia		
	At Poll	Absentee	Total	At Poll	Absentee	Total
Distance to Polling Place (mi)	-0.03 (0.31)	0.28*** (0.09)	0.25 (0.30)	-1.08*** (0.12)	1.34*** (0.12)	0.26** (0.13)
N	753,977	753,977	753,977	348,292	348,292	348,292
Outcome mean	52.93	1.74	54.67	24.70	27.68	52.38
R^2	0.11	0.03	0.12	0.05	0.17	0.20
<i>B. Border Fixed Effects</i>						
	Pennsylvania			Georgia		
	At Poll	Absentee	Total	At Poll	Absentee	Total
Distance to Polling Place (mi)	-0.64*** (0.23)	0.09* (0.05)	-0.55** (0.21)	-1.21*** (0.13)	1.35*** (0.14)	0.15 (0.15)
N	753,977	753,977	753,977	348,292	348,292	348,292
Outcome mean	52.93	1.74	54.67	24.70	27.68	52.38
R^2	0.12	0.03	0.13	0.05	0.18	0.21
<i>C. Matched Pair Fixed Effects</i>						
	Pennsylvania			Georgia		
	At Poll	Absentee	Total	At Poll	Absentee	Total
Distance to Polling Place (mi)	-0.55 (0.66)	0.05 (0.14)	-0.50 (0.63)	-1.34*** (0.43)	1.29** (0.51)	-0.05 (0.48)
N	1,104,290	1,104,290	1,104,290	548,398	548,398	548,398
Outcome mean	53.03	1.67	54.69	24.85	28.15	53.00
R^2	0.55	0.51	0.56	0.52	0.58	0.60

Note: This table reports estimates from linear fixed effects regressions. Regressions in Panel A include precinct fixed effects, regressions in Panel B include border segment fixed effects, and regressions in Panel C include matched-pair fixed effects. Each regression includes individual-level covariates (age, sex, registered democrat, registered republican), and block-level covariates (percent White, non-Hispanic, percent Black, non-Hispanic, percent Hispanic, median household income, percent with no high school degree, and percent that walk to work, cars per household). Outcome variables indicate whether a registered voter voted in person (at poll), by mail (absentee), or by either method (total). Outcomes are scaled so that coefficients are measured in percentage points. Distance (mi) measures the travel route distance between a voter and their assigned polling place. Standard errors allow for clustering at the precinct level in Panel A, at the border segment level in Panel B, and at the border segment and precinct levels in panel C.

5.3 Heterogeneous effects

In this section, we investigate differences in the responses to distance to polling place by demographic, economic, and political factors. To test for heterogeneous effects, we compare RD coefficients across sub-samples.³⁶ These results are suggestive: we do not have exogenous variation in the covariates of interest, but we can observe associations between covariates and estimated treatment effects. We first consider heterogeneous effects for covariates that we observe at the individual-level from voter registration files: age, sex, political party affiliation, and race/ethnicity (available for Georgia only). Using data from census block groups, we test for heterogeneous effects by educational attainment, income, and mode of transportation to work. Since we do not observe these characteristics at the individual level, we estimate effects for sub-samples of border segments with above and below median values of the continuous covariate.

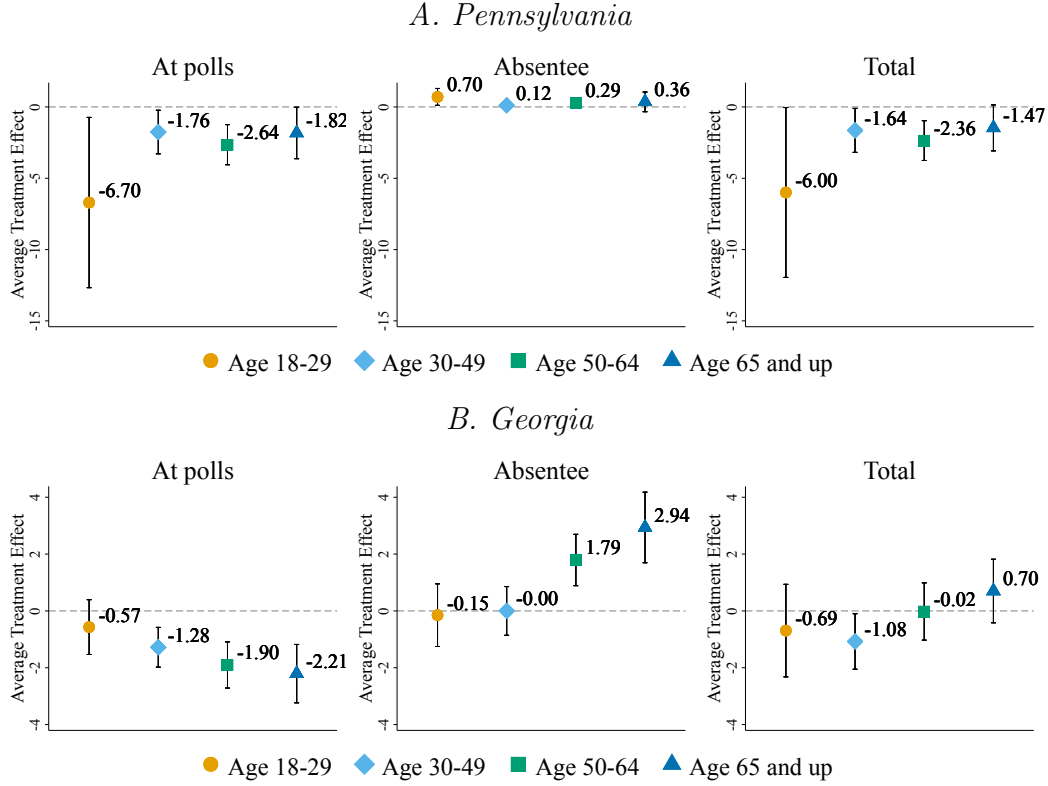
Age and Sex. One might expect younger voters to be more sensitive to costs of voting, if they have not yet developed a habit of voting (Fujiwara et al. 2016; Plutzer 2002). On the other hand, older voters may be more likely to request an absentee ballot as distance to polling place increases due to health and accessibility concerns. In Figure 5, point estimates suggest that younger voters are more sensitive to distance to polling place in Pennsylvania, though these differences are not statistically significant, and younger voters also tend to live closer to polling places (Table J.1). In Georgia, there is no indication that younger voters are more sensitive, and belonging to the age group 18-29 is also uncorrelated with distance to polling place in the RD sample. Instead, in Georgia, the estimated substitution to mail-in voting is larger among older voters than younger voters. In Pennsylvania, there is no significant substitution to mail in voting for any age group, even among voters aged 65 and above who are more likely to qualify for an absentee ballot. We find no differences in sensitivity to distance to polling place between male and female registered voters (Table J.3).

Race and Ethnicity. Race and ethnicity are especially important to consider in the context of the cost of voting due to both a persistent turnout gap between white and non-white voters (Fraga 2018; Ansolabehere et al. 2021) and long-standing concerns of voter disenfranchisement. During the 2018 elections, decisions to close or move polling places came under scrutiny due to concerns over voter suppression.³⁷ Related to these concerns, recent studies show that Black voters are more likely to experience longer waiting times at polls (Chen et al. 2020) and are more likely to have their mail-in ballots rejected (Shino et al. 2021).

³⁶We test if coefficients are equal across sub-samples using standard t-tests, with p-values reported in Tables in Appendix J.

³⁷“The Georgia Governor’s Race Has Brought Voter Suppression Into Full View”, *The Atlantic*. Retrieved November 6, 2018. “Republican Voter Suppression Efforts Are Targeting Minorities, Journalist Says”, *NPR*. Retrieved October 23, 2018.

Figure 5: Heterogeneous Effects by Age

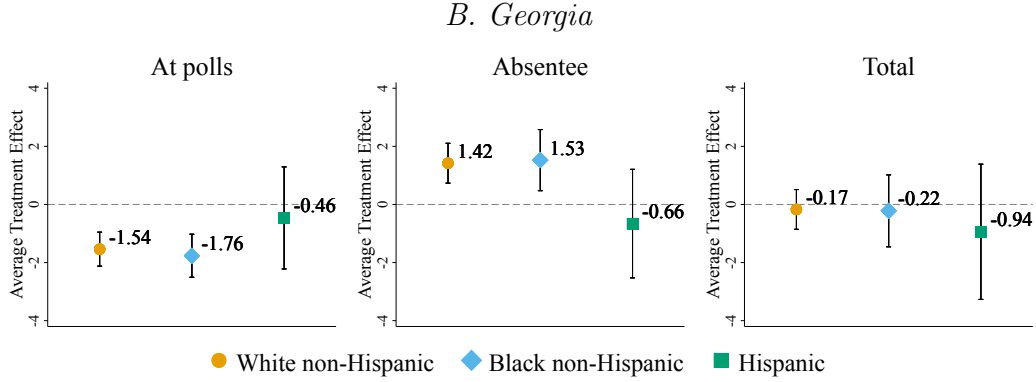


Note: This figure shows RD estimates for sub-samples of registered voters by age. The y-axis is measured in percentage points of likelihood of voting per mile of distance to polling place. Vertical lines indicate 95% confidence intervals. Standard errors allow for clustering at the RD border segment level.

In Figure 6, we find no statistically significant differences in how the overall likelihood of voting responds to distance to polling place between white non-Hispanic, Black non-Hispanic, and Hispanic voters in Georgia. This is despite the fact that Black non-Hispanic voters tend to live closer to the polling place than other voters (Table J.1), and voters who live close to the polling place tend to be more sensitive (Figure 4). Despite the similar effects of distance to polling place on overall turnout across groups, estimates suggest differential rates of substitution to voting by mail. The increase in absentee voting per mile increase in distance to polling place is statistically significantly lower among Hispanic voters relative to white non-Hispanic voters and to black non-Hispanic voters. Existing evidence of differential effects of distance to polling place by race is mixed. Cantoni (2020) finds that areas with a larger non-white population are more sensitive to distance to polling place, while Clinton et al. (2020) find that non-white voters are less likely to substitute to early voting in response to polling place changes than white voters. In studying recent polling place and precincting decisions

in North Carolina, Shepherd et al. (2021) find no evidence of manipulation of polling place choices that would systematically affect voters differently by race. The mixed findings suggest that context-specific factors and electoral design are important to take into account when considering whether or not election rule will have disparate racial impacts.

Figure 6: Heterogeneous Effects by Race and Ethnicity

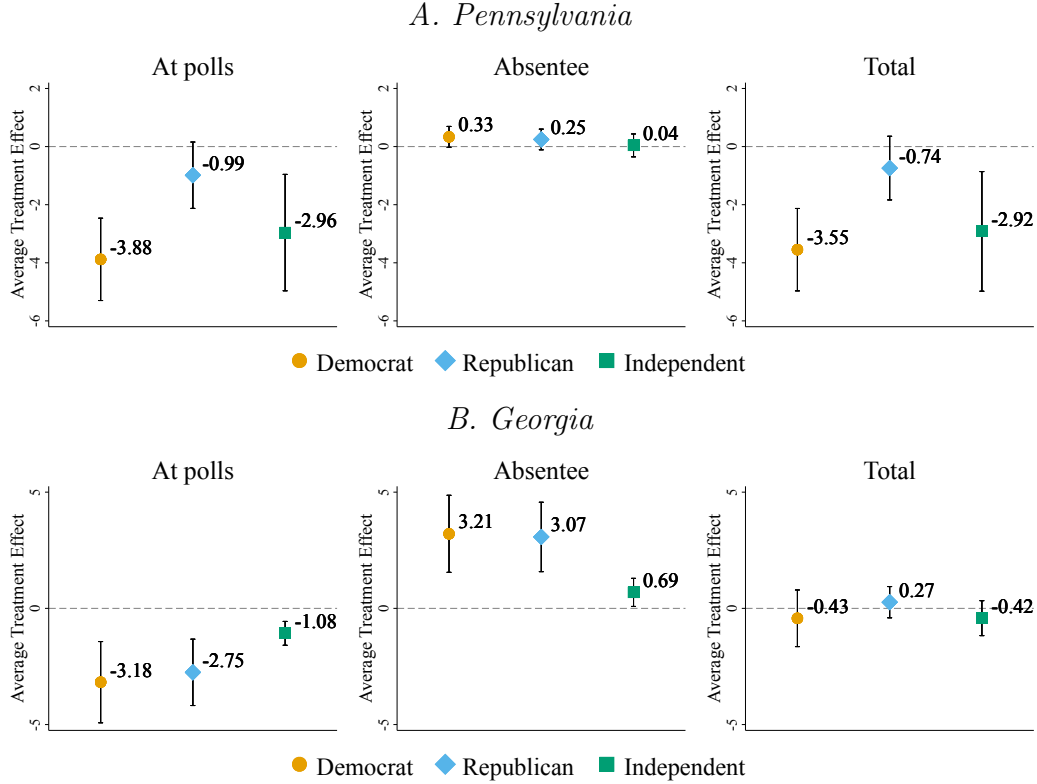


Note: This figure shows RD estimates for sub-samples of registered voters by race and ethnicity, as reported on voter registration files. This information is available at the individual-level in Georgia only. The y-axis is measured in percentage points of likelihood of voting per mile of distance to polling place. Vertical lines indicate 95% confidence intervals. Standard errors allow for clustering at the RD border segment level.

Political Party Affiliation. Using party affiliation from voter registration files, we estimate the effect separately for registered Democrats, registered Republicans, and for voters that do not register with either party (Figure 7). Estimates suggest that Republicans are less sensitive to changes in distance to polling place than Democrats in Pennsylvania and that Independents are less sensitive than both Democrats and Republicans in Georgia. However, these differences are difficult to disentangle from the nonlinear effects in Figure 4, since Democrats tend to live closer to their polling place than Republicans or Independents in both states (Table J.1). Further, Georgia and Pennsylvania are both swing states, so these findings may not generalize to states with strong support for one political party.

Income and Education. Participation in elections in the United States has historically been higher among those with relatively high income and educational attainment (Milligan et al. 2004, Sondheimer and Green 2010, Leighley and Nagler 2014). In theory, the direction of how income might moderate the effect of distance to polling place on turnout is unclear. On one hand, high-income voters may have a higher opportunity cost of voting because of their income-earning opportunities. On the other hand, low-income voters may have a higher opportunity cost of voting due to lower flexibility to take time out of work and because they are more likely to receive hourly

Figure 7: Heterogeneous Effects by Political Party

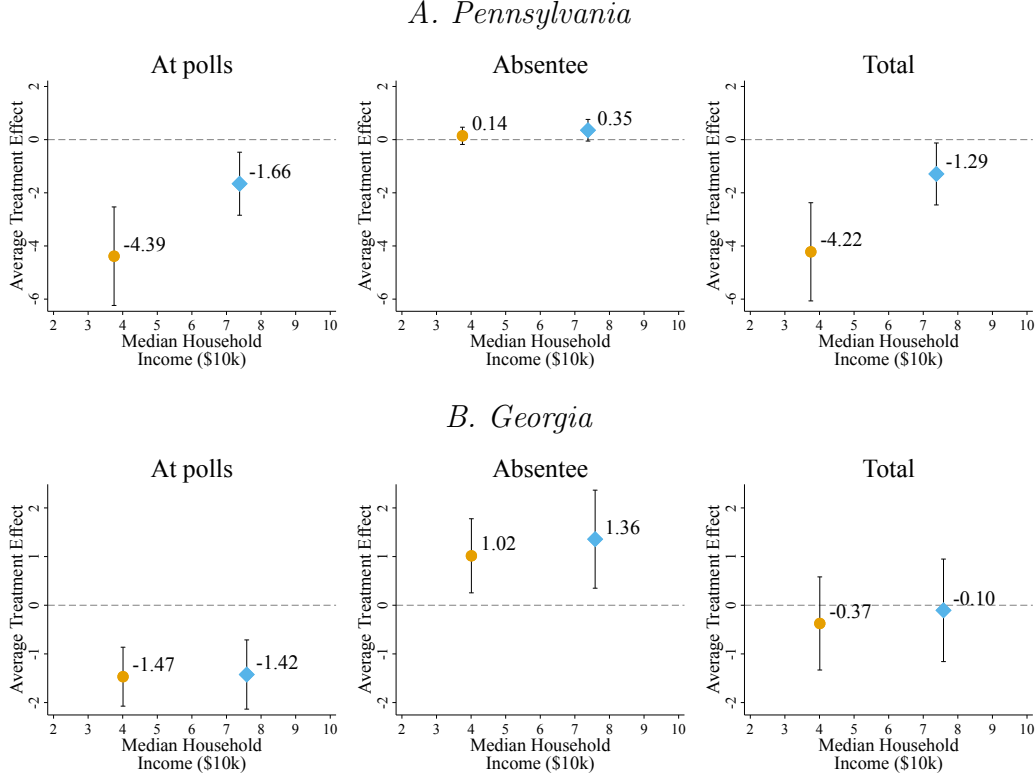


Note: This figure shows RD estimates for sub-samples of registered voters by party affiliation, as reported on voter registration files. The y-axis is measured in percentage points of likelihood of voting per mile of distance to polling place. Vertical lines indicate 95% confidence intervals. Standard errors allow for clustering at the RD border segment level.

wages. In both states, voters in relatively low income and low educational attainment areas tend to live closer to the polling place. In Figure 8, the likelihood of voting in person is more sensitive to distance to polling place in relatively low income areas in Pennsylvania, though there are no statistically significant differences between low and high income areas in Georgia. Patterns are similar for educational attainment (Table J.7).

Transportation. Finally, the cost of distance to the polling place will vary by mode of transportation. We have already seen that voters who live closer to their polling place are more sensitive to distance to polling place in Section 5.1. Voters who live closer to their polling place are also less likely to drive to the polls. It could be that nonlinear effects in Figure 4 represent associations between distance to polling place and time to polling place. We test this hypothesis by constructing a measure of time to polling place that takes mode of transportation into account. We assume that voters within 0.5 miles of the polling place walk along the travel route at 3 miles per hour, and

Figure 8: Heterogeneous Effects by Income



Note: This figure shows RD estimates for sub-samples of border segments with above and below median income. The income-level of border segment is measured as the average of the median household income. The y-axis is measured in percentage points of likelihood of voting per mile of distance to polling place. The x-axis shows the average median household income of each sub-sample. Vertical lines indicate 95% confidence intervals. Standard errors allow for clustering at the border segment level.

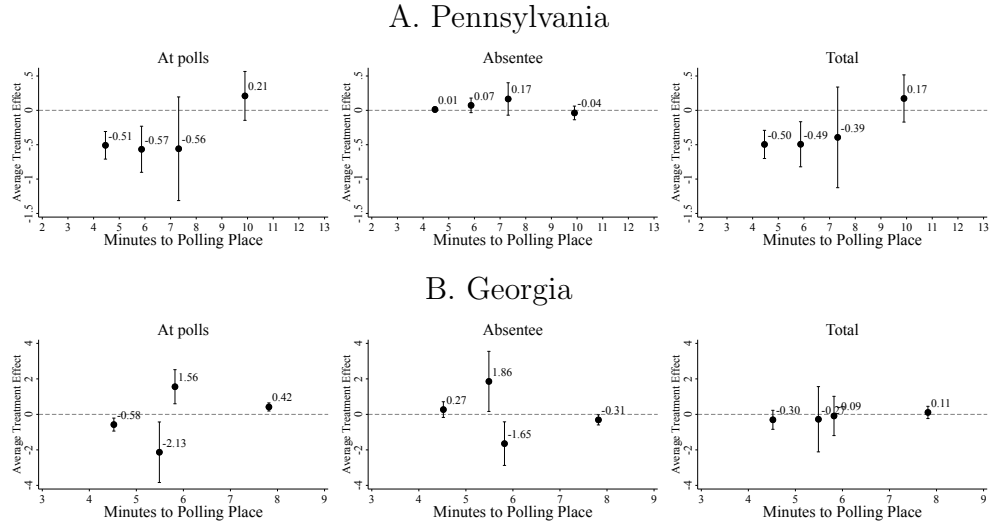
that all other voters drive along the travel route according to the speed limit.³⁸ Figure 9 shows that, using this measure of time to polling place, there is no strong evidence of non-linear effects. Although there is no way to know the actual mode of transportation and travel time experienced by voters, we interpret this as suggestive evidence that the non-linear effects in Figure 4 reflect differences in mode of transportation and travel time costs.

To further test this hypothesis, we estimate effects for sub-samples of border segments in areas where there is an above and below median rate of commuters who travel to work by walking. In Table J.8, we see that voters in areas with above median rates of commuting by walking are more sensitive to distance to polling place in Pennsylvania. The effect of an additional mile to polling place reduces the likelihood of voting in person by 4.2 p.p. (SE=1.0) in areas with high rates of commuting by foot (10% com-

³⁸Three miles per hour is the average walking speed used by Google Maps.

mute by walking, on average), versus only 1.9 p.p. (SE=0.6) in areas with low rates of commuting by foot. In Georgia, there are no significant differences, which may be due to the lower rates of people who commute by walking overall (3% commute by walking in the above-median sample). There are no significant differences in substitution to voting by mail by mode of transportation in either state. This is consistent with the idea that transportation affects the cost of voting in person, but not the costs of voting by mail.

Figure 9: Average treatment effects by travel time to polling place, assuming that those within 0.5 miles walk to the polls and all others drive



Note: This figure shows binned RD estimates. The RD sample is divided into quartiles based on the average time to the polling place among observations in the control precinct of a border segment. To measure time to the polling place, we assume that voters within 0.5 miles of the polling place walk instead of drive. The y-axis is measured in percentage points of likelihood of voting. The x-axis shows the average distance to polling place within each bin. Vertical lines indicate 95% confidence intervals. Standard errors allow for clustering at the border segment level.

5.4 Voter registration costs and search costs

In this section we address additional costs of voting that are related to polling place locations. First, a larger distance to polling place may reduce the likelihood that a voter registers to vote if the voter is aware of the polling place location in advance. We test for voter registration effects using a block-level border fixed effects regression.³⁹ At the block level, we have information on the voting age population, a proxy for voting eligibility, from the 2010 Census. We regress the percent of the block's voting

³⁹Cantoni (2020) similarly aggregates data to the block-level, noting that the benefit of avoiding selection bias comes at the cost of less precise measurement of distance.

age population that is registered to vote on the distance from the block centroid to the polling place. We include border segment fixed effects, and only include blocks that contain voters within 0.05 miles of the border. We find no evidence of a change in voter registration rates due to increased distance to polling place (Table F.1). The point estimates are small and statistically insignificant: they imply that a mile increase in distance to polling place is associated with a 0.1 and 0.5 percentage point reduction in registration rates in Pennsylvania and Georgia, respectively. Consistent with there being no large effects on voter registration, we find point estimates similar to those in Table 2 when we regress turnout (number of votes per voting age population) on distance to polling place (Table F.1).

Second, changes to distance in polling place in practice might be accompanied by additional short-term costs for a voter that needs to locate the new polling place, either because the polling place has moved or because the voter has moved to a new location. Such search costs are the focus of Clinton et al. (2020) and Yoder (2018). We construct a panel of voters and polling place locations for Pennsylvania (from 2016 to 2018). Using a differences-in-differences approach, we find that if a polling place moves, there is no statistically significant effect on the likelihood of voting. But, if a voter moved to a new election precinct, then there is an estimated 3.0 p.p. reduction in the likelihood of voting at polls in the 2018 general election (Online Appendix I).

6 Turnout-maximizing Polling Place Locations

In the previous empirical sections, we estimate the effect of a one-mile increase in distance to polling place on the likelihood of voting in person, by mail, and overall. From these estimates alone, it is difficult to predict how polling place location choices influence aggregate turnout in an election. For any change in polling place location, the distance to polling place might increase for some voters and decrease for others. With the goal of understanding how polling place changes affect voter participation in the aggregate, we use the RD estimates to find turnout-maximizing polling place locations for each precinct.⁴⁰ Since the effect of distance to polling place on the overall likelihood of voting is small and statistically insignificant in Georgia, we do this exercise for Pennsylvania only.

We explore three scenarios for counterfactual polling place locations. First, we find

⁴⁰The turnout-maximizing set of polling places is the relevant counterfactual for analyzing changes in voter participation. This objective is also aligned with the stated objective of local election officials who choose where to put polling places. Other objectives, like maximizing partisan advantage, could also be interesting, though we find no strong evidence of heterogeneous effects by partisanship in Section 5.3. Further, Ferrer et al. (2023) find no effect of the partisan make-up of a county election commission on the county’s turnout or vote shares.

the turnout-maximizing latitude and longitude for a polling place in each precinct. In the second exercise, we take existing infrastructure into account and identify the turnout-maximizing location among existing public buildings in a precinct.⁴¹ Third, we consider the experiment of doubling the number of polling locations per precinct, and identify the two turnout-maximizing public buildings per precinct. For these counterfactual exercises, we use the nonlinear RD estimates from Figure 4 and a linear probability model of vote choice.⁴² For the first exercise, we numerically solve for the turnout-maximizing latitude and longitude for a given allocation of voters. This is an unconstrained problem, in that the optimal polling location may not be near any existing infrastructure and may even fall outside the boundaries of the precinct. For the second and third exercise, we select the building or pair of buildings that maximize the gains to turnout from the existing set of possible buildings in a precinct. See Online Appendix K for more details.

Figure 10 shows the locations of existing and optimal polling places for all precincts in Pennsylvania, as well as for a comparative urban and rural area within the same county. This rough look at the raw data and our computed optimal polling places shows that the counterfactual locations are not too far from existing polling places. However, for 36% of precincts, the optimal polling place location is located outside the boundaries of the precinct. This suggests that there may be gains to re-precincting, that is, to re-draw precincts so that they include the optimal polling place.⁴³

The distribution of gains in turnout from implementing turnout-maximizing polling places are shown in Figure 11. The optimal polling place location tends to be relatively close to the existing polling place, with a median distance of 0.4 miles. The resulting median increase in turnout is approximately 100 votes per precinct, equivalent to an 11 p.p. or 22% increase in turnout.⁴⁴

⁴¹We construct a new dataset of polling place candidates by combining the locations of all public schools, libraries, court houses, police stations, and places of worship. These data are available from the Homeland Infrastructure Foundation-Level Data (HIFLD) of the Geospatial Management Office of the U.S. Department of Homeland Security, available at: <https://hifld-geoplatform.opendata.arcgis.com>. We also include the existing polling place as a candidate.

⁴²Specifically, we use the the quartile from Figure 4 that the voter belongs to under the existing polling place location. Then, for each voter, we draw an individual-specific marginal effect from a Normal distribution with the mean and standard deviation matching the point estimate and standard error for that bin. Finally, the change in the likelihood of voting for each individual and counterfactual polling place is the individual’s marginal effect multiplied by the change in distance.

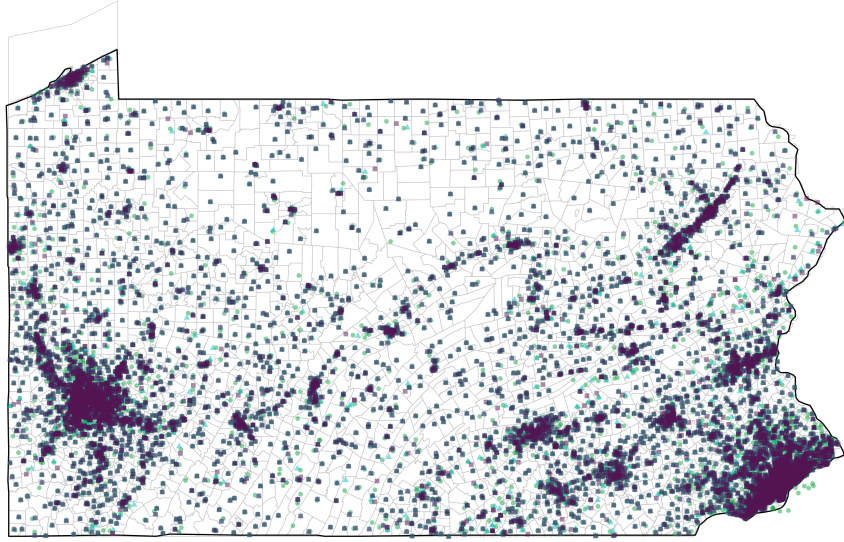
⁴³Precinct boundaries change relatively infrequently, typically as part of a decennial redistricting process. Polling places can change with each election cycle. Cantoni (2020) considers an optimal re-precincting problem, given existing locations of polling places. The objective is to minimize distance, whereas we consider a turnout-maximization problem, allowing for nonlinear effects of distance on turnout.

⁴⁴We measure turnout as the number of votes per voting age population in order to compare our results with other studies of voter participation. By choosing VAP instead of registered voters in the denominator, we assume that no unregistered voters respond to changes in distance to polling place, consistent with the findings in Section 5.4.

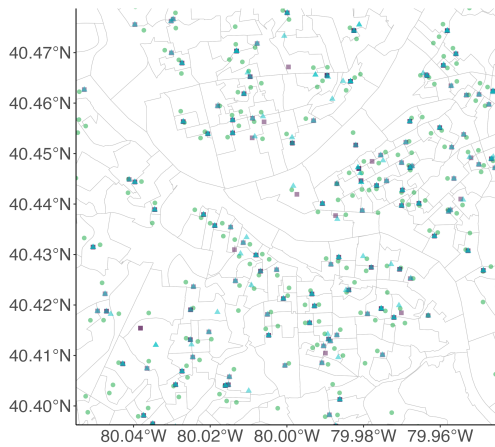
Figure 10: Optimal and Existing polling place locations in Pennsylvania

(a) All of Pennsylvania

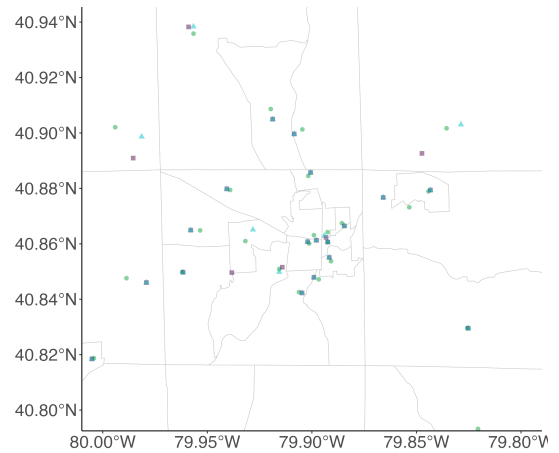
■ Current Polling Place ● Optimal Polling Place ▲ Optimal Public Building



(b) Urban area in Allegheny County, PA



(c) Rural area in Allegheny County, PA



Note: This figure shows the locations of optimal and existing polling places. Panel A shows a map of the state of Pennsylvania, with current polling locations indicated in red and optimal polling places indicated in blue. The gray lines show precinct borders.

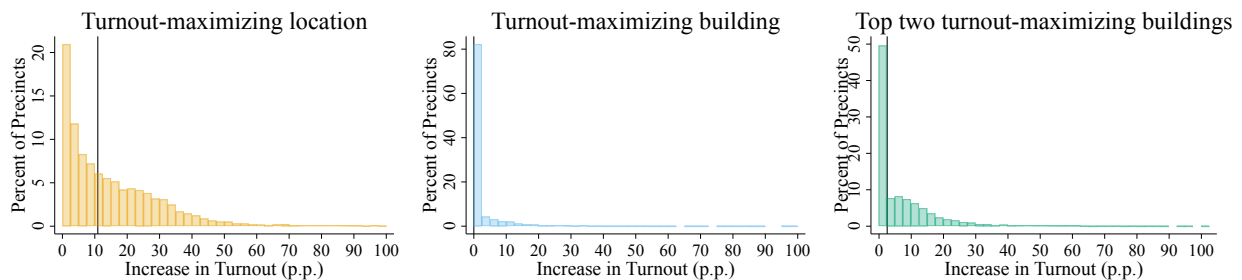
The gains to voter participation from this first exercise may be unrealistic, since many precincts have only a handful of buildings that would be suitable polling places. The number of polling place candidates per precinct ranges from 1 to 19, with a median value of 2. In the second counterfactual exercise, we identify the turnout-maximizing building, among the known candidates. The median increase in turnout per precinct is zero, as the existing polling place is also the turnout-maximizing building for roughly 70% of precincts. However, among the remaining 30% of precincts, the median increase in turnout is 5 p.p., a 10% increase. Statewide, if Pennsylvania were to use all turnout-maximizing buildings as polling places, there would be a gain of approximately 290,000 votes, equivalent to a 3 p.p. increase in turnout (6%).

For perspective, these gains are on par with other relatively successful voter mobilization tactics and policies. In a large-scale field experiment in the U.S., mailings that add elements of social pressure (e.g., the voting history of neighbors) increase turnout by 2 p.p. (Gerber et al. 2017). The average effect of canvassing across 71 studies is 3 p.p. (Green et al. 2013). Kaplan and Yuan (2020) estimate that an additional day of early voting increases turnout by 0.2 percentage points. This makes the statewide effect equivalent to about 2 weeks of early voting.

Finally, we consider doubling the number of polling places, using existing buildings. Approximately 30% of precincts have only one candidate building and hence no possible gains under this counterfactual exercise. Nonetheless, the median increase in turnout is 20 votes per precinct, a 3 p.p. increase (6%). If Pennsylvania were to double the number of polling places statewide, where feasible, then the overall gain in turnout would be 895,000 votes, a 9 p.p. increase in turnout (18%). This is on par with the estimated effect of large scale presidential campaigns in the most heavily targeted states, including all advertising and canvassing (Enos and Fowler 2018).

While doubling the number of polling places could be costly, we can also use counterfactual simulations to identify a small number of precincts with large gains to re-optimizing existing polling place locations. The 90th percentile increase in turnout from simply re-locating the polling place to another existing building is 8 p.p.. Compared to other voter mobilization efforts, relocating or increasing the number of polling places might be a cost-effective way to improve voter participation.

Figure 11: Distributions of difference in turnout between turnout-maximizing polling place and existing polling place



Note: For each simulation exercise, this figure shows the distribution of the difference in turnout between the optimal and existing polling place allocations at the precinct-level. Difference in turnout is measured as the change in the number of votes per voting age population (p.p.). Median values are indicated by vertical lines. To enhance readability, the histograms omit observations with greater than 100 p.p. increase in turnout. This excludes 1%, 0.3%, and 0.2% of precincts for the three counterfactual exercises, respectively.

7 Conclusion

We study the causal effect of distance to polling place on voter participation and voting method, at polls or by mail, in two large swing states. On average, there is a small negative effect of distance to polling place on the likelihood that a registered voter goes to the poll to vote. A standard deviation increase in distance to polling place reduces the likelihood of voting in person by 1.4 and 1.6 percentage points in Pennsylvania and Georgia, respectively. We find that effects are non-linear, with voters closest to polling places being most sensitive to an increase in distance to the polling place, likely due to differences in mode of transportation. In Georgia, voters substitute to voting by mail, such that there is no net effect of distance to polling place on the likelihood of voting. In Pennsylvania, substitution to mail-in voting is limited and there is a negative net effect of distance to polling place on the likelihood of voting. An important difference between the states in the 2018 elections is that voters needed no excuse to vote by mail in Georgia and did need an excuse to vote by mail in Pennsylvania.

In our preferred specification, we adapt a geographic RD methodology for the case of a continuous treatment variable, where the change in treatment is not necessarily monotonic at the discontinuity. Our application is useful for future studies of voting costs, which might similarly exploit election precinct boundaries in other settings, and for research involving geographic discontinuities more broadly.

The results highlight some important lessons for studies of electoral design and voter participation in the future. First, it is important to use large datasets to study costs of voting in large elections, where power is needed to detect very small effects.

The ability to estimate small effects with precision is especially important in settings where we expect close margins of victory. Second, the comparison of our findings to those in the existing literature as well as our own heterogeneous effects analysis suggest that changes in voting behavior can be highly context-specific. Third, it is important to take context into account when determining the effect of any election design change. For example, the effect of changes to polling place locations likely depends on the availability of voting by mail and other convenience voting methods. Cantoni and Pons (2022) show that a voter’s context and location influences whether or not they vote. This paper adds that voters’ sensitivity to costs of voting is also highly context-dependent.

The findings in this paper can help election commissions that face costly trade-offs in choosing how many polling places to open and where to place them. Our counterfactual exercise shows that the causal estimates from our paper are not merely of academic interest but are policy-relevant if local officials want a practical way of increasing turnout. In Pennsylvania, using the set of turnout-maximizing public buildings for polling places could increase turnout by 3 percentage points in aggregate. In future work, one might take into account the differential costs of voting in person and by mail and of opening and closing polling places, in order to determine the cost efficiency of implementing turnout-maximizing polling place locations. With a more nuanced understanding of when voters choose to vote in person, vote by mail, or abstain, election commissions have an opportunity to reduce costs of voting or to make costs of voting more equal across the population.

References

- Alipour, J.-V. and V. Lindlacher (2023). No surprises, please: Voting costs and electoral turnout. *CESifo Working Paper, No. 9759*.
- Amos, B., D. A. Smith, and C. Ste. Claire (2017). Reprecincting and Voting Behavior. *Political Behavior* 39(1), 133–156.
- Angrist, J. D. and J.-S. Pischke (2009). *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton university press.
- Ansolabehere, S., B. L. Fraga, and B. F. Schaffner (2021). The CPS Voting and Registration Supplement Overstates Minority Turnout. *Journal of Politics*, Forthcoming.
- Avery, J. M. (2015). Does Who Votes Matter? Income Bias in Voter Turnout and Economic Inequality in the American States from 1980 to 2010. *Political Behavior* 37(4), 955–976.
- Bentele, K. G. and E. E. O’Brien (2013). Jim crow 2.0? why states consider and adopt restrictive voter access policies. *Perspectives on Politics* 11(4), 1088–1116.
- Braconnier, C., J. Y. Dormagen, and V. Pons (2017). Voter Registration Costs and Disenfranchisement: Experimental Evidence from France. *American Political Science Review* 111(3), 584–604.
- Brady, H. E. and J. E. McNulty (2011). Turning out to Vote: The Costs of Finding and Getting to the Polling Place. *American Political Science Review* 105(1), 115–134.
- Burden, B. C., D. T. Canon, K. R. Mayer, and D. P. Moynihan (2017). The complicated partisan effects of state election laws. *Political Research Quarterly* 70(3), 564–576.
- Cantoni, E. (2020). A Precinct Too Far: Turnout and Voting Costs. *American Economic Journal: Applied Economics* 12(1), 61–85.
- Cantoni, E. and V. Pons (2021). Strict id laws don’t stop voters: Evidence from a u.s. nationwide panel, 2008–2018. *Quarterly Journal of Economics* 136(4), 2615–60.
- Cantoni, E. and V. Pons (2022). Does context outweigh individual characteristics in driving voting behavior? evidence from relocations within the united states. *American Economic Review* 112(4), 1226–72.
- Card, D., D. S. Lee, Z. Pei, and A. Weber (2015). Inference on causal effects in a generalized regression kink design. *Econometrica* 83(6), 2453–2483.

- Cattaneo, M. D., B. R. Frandsen, and R. Titiunik (2015). Randomization inference in the regression discontinuity design: An application to party advantages in the us senate. *Journal of Causal Inference* 3(1), 1–24.
- Cattaneo, M. D., M. Jansson, and X. Ma (2020). Simple local polynomial density estimators. *Journal of the American Statistical Association* 115(531), 1449–1455.
- Chen, M. K., K. Haggag, D. G. Pope, and R. Rohla (2020). Racial Disparities in Voting Wait Times: Evidence from Smartphone Data. *Review of Economics and Statistics*, Forthcoming.
- Clinton, J. D., N. Eubank, A. Fresh, and M. E. Shepherd (2020). Polling Place Changes and Political Participation: Evidence from North Carolina Presidential Elections, 2008-2016 *. *Political Science Research and Methods First View*, 1–18.
- Dell, M. (2010). The persistent effects of peru’s mining mita. *Econometrica* 78(6), 1863–1903.
- Dube, A., T. W. Lester, and M. Reich (2010). Minimum wage effects across state borders: Estimates using contiguous counties. *The Review of Economics and Statistics* 92(4), 945–964.
- Dyck, J. J. and J. G. Gimpel (2005). Distance, turnout, and the convenience of voting. *Social Science Quarterly* 86(3), 531–548.
- Enos, R. D. and A. Fowler (2018). Aggregate effects of large-scale campaigns on voter turnout. *Political Science Research and Methods* 6(4), 733–751.
- Ferrer, J., I. Geyn, and D. M. Thompson (2023). How partisan is local election administration? *American Political Science Review*, 1–16.
- Fraga, B. L. (2018). *The Turnout Gap*. Cambridge University Press.
- Fujiwara, T., K. Meng, and T. Vogl (2016). Habit formation in voting: Evidence from rainy elections. *American Economic Journal: Applied Economics* 8(4), 160–188.
- Gerber, A. S., G. A. Huber, A. H. Fang, and A. Gooch (2017). The generalizability of social pressure effects on turnout across high-salience electoral contexts: Field experimental evidence from 1.96 million citizens in 17 states. *American Politics Research* 45(4), 533–559.
- Gomez, B. T., T. G. Hansford, and G. A. Krause (2007). The republicans should pray for rain: Weather, turnout, and voting in U.S. presidential elections. *Journal of Politics* 69(3), 649–663.

- Green, D. P., M. C. McGrath, and P. M. Aronow (2013). Field experiments and the study of voter turnout. *Journal of Elections, Public Opinion and Parties* 23(1), 27–48.
- Hasen, R. L. (2012). *The Voting Wars*. Yale University Press.
- Haspel, M. and H. Gibbs Knotts (2005). Location, location, location: Precinct placement and the costs of voting. *Journal of Politics* 67(2), 560–573.
- Highton, B. (2017). Voter identification laws and turnout in the united states. *Annual Review of Political Science* 20(1), 149–167.
- Jones, D., D. Molitor, and J. Reif (2019). What do workplace wellness programs do? evidence from the illinois workplace wellness study. *The Quarterly Journal of Economics* 134(4), 1747–1791.
- Kaplan, E. and H. Yuan (2020). Early voting laws, voter turnout, and Partisan vote composition: Evidence from Ohio. *American Economic Journal: Applied Economics* 12(1), 32–60.
- Keele, L. J. and R. Titiunik (2015). Geographic boundaries as regression discontinuities. *Political Analysis* 23(1), 127–155.
- Leighley, J. E. and J. Nagler (2014). *Who votes now?* Princeton University Press.
- Lockhart, M., S. J. Hill, J. Merolla, M. Romero, and T. Kousser (2020). America’s electorate is increasingly polarized along partisan lines about voting by mail during the covid-19 crisis. *Proceedings of the National Academy of Sciences* 117(40), 24640–24642.
- Martin, P. S. (2003). Voting’s rewards: Voter turnout, attentive publics, and congressional allocation of federal money. *American Journal of Political Science* 47(1), 110–127.
- Meredith, M. and Z. Endter (2016). Aging into absentee voting: Evidence from texas. *Working Paper*.
- Meredith, M. and N. Malhotra (2011). Convenience voting can affect election outcomes. *Election Law Journal* 10(3), 227–253.
- Milligan, K., E. Moretti, and P. Oreopoulos (2004). Does education improve citizenship? evidence from the united states and the united kingdom. *Journal of Public Economics* 88(9), 1667 – 1695.

- Plutzer, E. (2002). Becoming a Habitual Voter: Inertia, Resources, and Growth in Young Adulthood. *The American Political Science Review* 96(1), 41–56.
- Shepherd, M. E., A. Fresh, N. Eubank, and J. D. Clinton (2021). The Politics of Locating Polling Places: Race and Partisanship in North Carolina Election Administration, 2008–2016. *Election Law Journal: Rules, Politics, and Policy* 20(2), 155–177.
- Shino, E., M. Suttman-Lea, and D. A. Smith (2021). Determinants of rejected mail ballots in georgia’s 2018 general election. *Political Research Quarterly* 75(1), 231–243.
- Sondheimer, R. M. and D. P. Green (2010). Using experiments to estimate the effects of education on voter turnout. *American Journal of Political Science* 54(1), 174–189.
- Thompson, D. M., J. A. Wu, J. Yoder, and A. B. Hall (2020). Universal vote-by-mail has no impact on partisan turnout or vote share. *Proceedings of the National Academy of Sciences of the United States of America* 117(25), 14052–14056.
- Tomkins, S., K. Yao, J. Gaebler, T. Konitzer, D. Rothschild, M. Meredith, and S. Goel (2023). Blocks as geographic discontinuities: The effect of polling-place assignment on voting. *Political Analysis* 31(2), 165–180.
- Wolfinger, N. and W. Raymond (2008). Family Structure and Voter Turnout. *Social Forces* 86(5), 1863–1903.
- Yoder, J. (2018). How Polling Place Changes Reduce Turnout: Evidence from Administrative Data in North Carolina. *Working Paper*.

A Appendix

A.1 Summary Statistics

Table A.1: Summary Statistics

	Pennsylvania				Georgia			
	All		RD sample		All		RD sample	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
<i>Voting History</i>								
Primary election, at polls	19.14	39.34	19.12	39.32	12.13	32.64	12.12	32.64
Primary election, absentee	0.41	6.40	0.41	6.40	4.35	20.40	4.34	20.37
Primary election, total	19.55	39.66	19.53	39.64	16.48	37.10	16.46	37.09
General election, at polls	57.32	49.46	57.33	49.46	25.75	43.73	25.74	43.72
General election, absentee	2.28	14.92	2.28	14.93	28.93	45.35	28.85	45.31
General election, total	59.59	49.07	59.61	49.07	54.69	49.78	54.59	49.79
<i>Distance to polling place</i>								
Travel route distance (mi)	0.89	0.91	0.87	0.87	1.91	1.44	1.90	1.42
Euclidean distance (mi)	0.58	0.62	0.56	0.58	1.23	1.02	1.22	1.00
Travel time (min)	2.59	1.86	2.55	1.77	4.16	2.24	4.15	2.23
<i>Demographics</i>								
Democrat	0.52	0.50	0.52	0.50	0.09	0.29	0.09	0.29
Republican	0.34	0.47	0.34	0.47	0.07	0.26	0.07	0.26
Independent	0.14	0.35	0.14	0.35	0.00	0.05	0.00	0.05
Percent Female	0.52	0.07	0.52	0.07	0.52	0.07	0.52	0.07
Percent urban	0.95	0.21	0.96	0.20	0.91	0.29	0.91	0.29
Percent Black	0.13	0.25	0.13	0.25	0.35	0.34	0.35	0.34
Percent White	0.79	0.28	0.79	0.28	0.55	0.34	0.55	0.34
Percent Hispanic	0.06	0.13	0.06	0.13	0.07	0.11	0.07	0.11
Median hh Income (10k USD)	5.93	3.09	5.93	3.09	6.10	3.07	6.10	3.07
Percent without a HS degree	0.11	0.10	0.11	0.10	0.13	0.11	0.13	0.11
Percent that walk to work	0.05	0.10	0.05	0.10	0.02	0.05	0.02	0.05
Cars per household	1.56	0.47	1.56	0.47	1.79	0.41	1.79	0.41
N	3,113,565		2,981,013		3,063,406		2,968,150	

Note: For each voting history variable, we observe whether or not a registered voter votes, by method of voting. We report these variables in percentage points to make regression coefficients easier to interpret. Demographic variables are measured at the block, or block-group level and assigned to each individual voter that resides in the geographic area. The RD sample is the Regression Discontinuity sample, described in Section 4.1.

A.2 Manipulation and Balance Tests for RD Specification

In this section, we estimate RD coefficients for covariates and test for manipulation of the running variable. First, we report RD estimates for twelve placebo outcomes in each state (Table A.2). To ensure that the samples are comparable, we use the same bandwidth as the one used for voting outcomes in Table 1. As with the main RD specification, we use residualized outcomes, after removing border segment fixed effects. We correct for multiple hypothesis testing with the Westfall-Young procedure, using Stata command *wyoung* by Jones et al. (2019) with 1,000 bootstrap replications.

Overall, we do not find strong evidence against the identifying assumption that potential confounding factors are continuous at the border. Estimated coefficients are small and most are statistically insignificant. There are three statistically significant coefficients (of 12) in Pennsylvania, and 1 in Georgia, which is perhaps not unlikely given the large sample size. In Pennsylvania, there is a statistically significant decrease in the probability a voter is female (-0.004 , $SE = 0.002$), decrease in age ($= 0.386$, $SE = 0.164$), and increase in cars per household (0.008 , $SE = 0.003$). These effects are small (the coefficients equal 0.9%, 0.8%, and 0.5% of the mean, respectively) and not statistically significant after correcting for multiple hypothesis testing. In Georgia, there is a statistically significant decrease in the likelihood of being in an urban block (-0.009 , $SE = 0.004$, equal to 1% of the mean). Similarly, this coefficient is not statistically significant after correcting for multiple hypothesis testing.

Next, we evaluate the distribution of the running variable near the discontinuity. Before evaluating continuity of the running variable at the border, we first note that the underlying distribution of the distance from voter to the border is irregular, primarily due to road patterns. Voters are not distributed evenly around precinct borders, but instead either live very close to the border, which is typically a road itself, or on the next nearest road. Figure A.1 shows the distance to the border for all voters within 0.5 miles of a border segment for Pennsylvania and Georgia. The distributions are bimodal. The first peak consists of voters that live on the road that acts as a border. Approximately 14% of the RD sample in Pennsylvania and 6% of the RD sample in Georgia live within 0.01 miles (about 50 feet or 16 meters) of the border segment. The second peak represents the modal distance to the next nearest road. The pattern is more pronounced in Pennsylvania, in part due to the grid pattern in Philadelphia, the largest city in either state.

Due to the shapes of these distributions, a standard McCrary test is difficult to implement. Nonetheless, for completeness, we use local polynomial density estimation to test for discontinuities in the running variable with the Stata package *rddensity*. Figure A.2 shows histograms of distributions of the running variable for each state. We show the distribution for 3 times the optimal bandwidth (the default) and use

Table A.2: RD Estimates for placebo outcomes

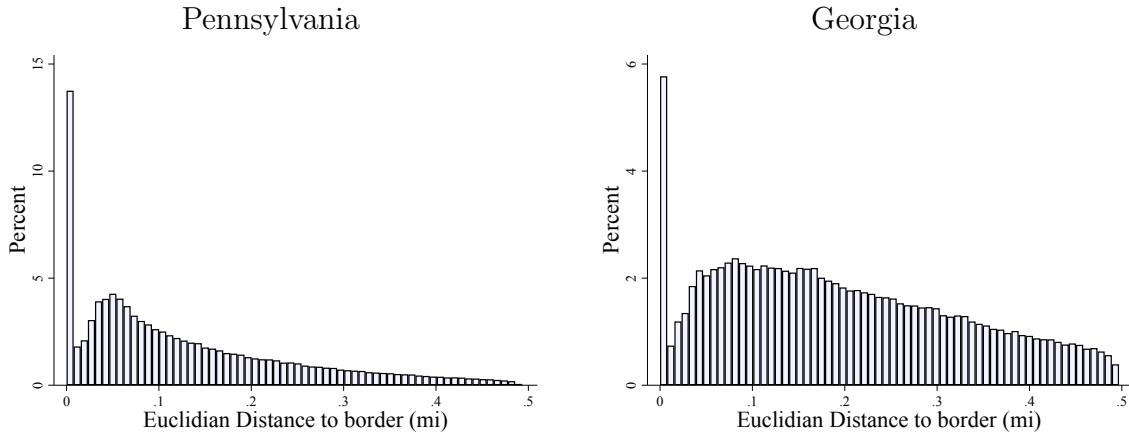
A. Pennsylvania						
	Female	Age	Democrat	Republican	Median HH Inc.	Urban
RD Estimate	-0.0042** (0.0018) [0.26]	-0.3856** (0.1639) [0.26]	-0.0006 (0.0019) [0.95]	-0.0017 (0.0019) [0.84]	0.0280 (0.0224) [0.79]	-0.0029 (0.0018) [0.61]
N	2,432,851	2,978,409	2,981,089	2,981,089	2,974,297	2,930,658
Effective N, Left	433,424	530,314	530,905	530,905	527,050	522,384
Effective N, Right	386,052	472,160	472,628	472,628	471,815	462,539
Bandwidth	0.06	0.06	0.06	0.06	0.06	0.06
Outcome mean	0.46	50.52	0.52	0.34	5.93	0.96
	Percent White	Percent Black	Percent Hispanic	Percent with No HS Degree	Percent that Walk to Work	Cars per Household
RD Estimate	-0.0031 (0.0022) [0.77]	0.0001 (0.0016) [0.95]	0.0006 (0.0012) [0.95]	-0.0010 (0.0011) [0.86]	0.0013 (0.0010) [0.79]	0.0077** (0.0032) [0.26]
N	2,930,658	2,930,658	2,930,658	2,975,723	2,980,814	2,529,423
Effective N, Left	522,384	522,384	522,384	527,360	530,900	464,013
Effective N, Right	462,539	462,539	462,539	471,858	472,614	408,786
Bandwidth	0.06	0.06	0.06	0.06	0.06	0.06
Outcome mean	0.79	0.13	0.06	0.11	0.05	1.56
B. Georgia						
	Female	Age	Democrat	Republican	Median HH Inc.	Urban
RD Estimate	-0.0026 (0.0031) [0.95]	-0.2105 (0.2536) [0.95]	-0.0011 (0.0021) [0.96]	-0.0029 (0.0019) [0.71]	0.0098 (0.0420) [0.96]	-0.0090** (0.0038) [0.20]
N	2,962,541	2,967,147	2,968,151	2,968,151	2,966,880	2,904,346
Effective N, Left	379,477	380,080	380,239	380,239	380,239	369,078
Effective N, Right	342,253	342,752	342,894	342,894	342,879	332,614
Bandwidth	0.09	0.09	0.09	0.09	0.09	0.09
Outcome mean	0.54	45.96	0.09	0.07	6.10	0.91
	Percent White	Percent Black	Percent Hispanic	Percent with No HS Degree	Percent that Walk to Work	Cars per Household
RD Estimate	0.0087 (0.0053) [0.71]	-0.0079 (0.0050) [0.71]	-0.0027 (0.0026) [0.93]	-0.0013 (0.0025) [0.96]	-0.0011 (0.0009) [0.93]	0.0042 (0.0074) [0.96]
N	2,904,346	2,904,346	2,904,346	2,968,150	2,968,150	2,610,017
Effective N, Left	369,078	369,078	369,078	380,239	380,239	338,389
Effective N, Right	332,614	332,614	332,614	342,894	342,894	304,784
Bandwidth	0.09	0.09	0.09	0.09	0.09	0.09
Outcome mean	0.55	0.35	0.07	0.13	0.02	1.79

Note: This table reports RD estimates for placebo outcomes. As in the main RD analysis, we use residualized outcomes, after removing border segment fixed effects. Female, Age, Democrat, and Republican are observed at the individual-level from voter registration files. Median household (HH) income, urban, race/ethnicity, percent with no high school (HS) degree, percent that walk to work, and cars per household are observed at the Census block or block-group level. Bandwidths are MSE-optimal. Standard errors allow for clustering at the border-segment level.

a fourth-order polynomial to estimate the density function. Despite this, the local polynomial density estimation does not match the distribution well. Point estimates for the jump in density at the discontinuity fall outside of bias-corrected confidence intervals proposed by Cattaneo et al. (2020). With these caveats, for both states, we reject the null hypothesis that there is no discontinuity at the cutoff (with p -values < 0.01).

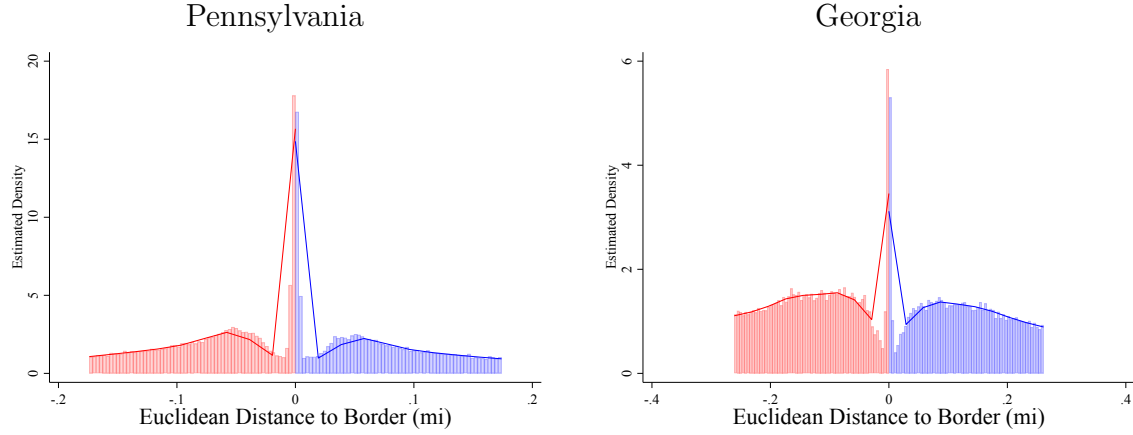
Given the irregular distribution of the running variable, we prefer the randomization inference approach of Cattaneo et al. 2015 in this setting. We test if the running variable is as good as randomly assigned for a set of increasingly large windows around the cutoff. The results of the binomial tests are reported in Table A.4. We follow the recommendation to include approximately 20 observations in the first window, and to expand the window in equal-sized increments for 10 windows. The smallest p -value is 0.05 in Pennsylvania and 0.57 in Georgia. If we consider the mass of voters that are within 0.01 miles to the border in both states (a much larger window than used in the binomial test), 47% of observations ($N=446,663$) are in the treatment precinct in Pennsylvania and 48% of observations ($N=182,959$) are in the treatment precinct in Georgia. Taken together, the balance tests and binomial tests fail to show strong evidence of manipulation of the running variable. This suggests that we successfully eliminated any precinct borders that would induce systematic sorting by voters that might violate the continuity assumption (e.g., precinct borders that overlap with school district borders).

Figure A.1: Distribution of distance from voter to border segment



Note: These figures show histograms for the Euclidean distance in miles between the voter and the nearest border segment, for all voters within 0.5 miles of any border segment.

Figure A.2: Manipulation tests: Estimated of density of running variables near the border



Note: These figures show histograms and estimated kernel densities of the running variable, which is the Euclidean distance from the voter to the nearest border segment (mi). The values are negative for the control precinct (shaded in red) and positive for the treatment precinct (shaded in blue).

Table A.3: Manipulation Test: Local polynomial density estimation

	Pennsylvania	Georgia
Bandwidth	0.06	0.09
Polynomial for estimation	4	4
Effective N of obs., left	530,905	380,239
Effective N of obs., right	472,628	342,894
T-statistic for density test	-15.33	-17.00
<i>p</i> -value for density test	< .001	< .001

Table A.4: Binomial Tests

A. Pennsylvania			
Window Length /2 (mi)	N Obs. Left of cutoff	N Obs. Right of cutoff	p -value for H_0 : prob. of left side = 0.5
0.0000003	16	7	0.09
0.0000007	21	12	0.16
0.0000010	23	18	0.53
0.0000014	28	22	0.48
0.0000017	31	27	0.69
0.0000021	34	29	0.61
0.0000024	38	37	1.00
0.0000028	45	44	0.92
0.0000031	51	44	0.54
0.0000034	60	49	0.34

B. Georgia			
Window Length /2 (mi)	N Obs. Left of cutoff	N Obs. Right of cutoff	p -value for H_0 : prob. of left side = 0.5
0.0000018	14	7	0.19
0.000036	18	17	1.00
0.000053	40	53	0.21
0.000071	56	65	0.47
0.000089	73	80	0.63
0.000107	89	89	1.00
0.000124	98	107	0.58
0.000142	118	133	0.38
0.000160	125	149	0.16
0.000178	134	164	0.09

Note: The p -value is for the null hypothesis that the probability of being on left of the threshold is 0.5.

A.3 Balance for Fixed Effects Regression Specifications

Table A.5: Correlations between covariates and distance to polling place, including border fixed effects

A. Pennsylvania						
	Female	Age	Democrat	Republican	Median HH Income	Urban
Distance to polling place (mi)	-0.0023*** (0.0008)	-0.1862* (0.1056)	-0.0025 (0.0017)	0.0022 (0.0017)	0.0141 (0.0275)	-0.0136*** (0.0030)
N	867,036	882,105	883,056	883,056	878,404	867,036
Outcome mean	0.52	49.12	0.61	0.25	5.02	0.97
R^2	0.35	0.15	0.16	0.17	0.90	0.79
	Percent White	Percent Black	Percent Hispanic	Percent with No HS Degree	Percent that Walk to Work	Cars per Household
Distance to polling place (mi)	0.0012 (0.0022)	-0.0025 (0.0020)	-0.0011 (0.0007)	-0.0009 (0.0007)	-0.0005 (0.0005)	0.0032 (0.0051)
N	867,036	867,036	867,036	878,752	883,044	769,043
Outcome mean	0.67	0.23	0.08	0.14	0.07	1.34
R^2	0.92	0.94	0.85	0.85	0.90	0.93
B. Georgia						
	Female	Age	Democrat	Republican	Median HH Income	Urban
Distance to polling place (mi)	-0.0017*** (0.0006)	-0.1439* (0.0753)	-0.0021*** (0.0007)	-0.0003 (0.0009)	0.0327** (0.0166)	-0.0127*** (0.0023)
N	390,194	404,836	405,001	405,001	404,987	390,194
Outcome mean	0.52	46.06	0.09	0.07	5.77	0.85
R^2	0.48	0.10	0.06	0.09	0.87	0.87
	Percent White	Percent Black	Percent Hispanic	Percent with No HS Degree	Percent that Walk to Work	Cars per Household
Distance to polling place (mi)	0.0018 (0.0021)	-0.0014 (0.0020)	-0.0015 (0.0010)	-0.0020** (0.0008)	-0.0007* (0.0003)	0.0124*** (0.0028)
N	390,194	390,194	390,194	405,001	405,001	362,150
Outcome mean	0.56	0.35	0.07	0.14	0.02	1.74
R^2	0.82	0.84	0.63	0.83	0.82	0.87

Note: This table reports coefficients for the travel route distance to polling place (mi). The outcome variables are the individual-level and census block-level covariates listed in each column. Each regression includes border segment fixed effects. Standard errors allow for clustering at border segment level.

Table A.6: Correlations between covariates and distance to polling place, including matched pair fixed effects

A. Pennsylvania						
	Female	Age	Democrat	Republican	Median HH Income	Urban
Distance to polling place (mi)	-0.005*** (0.001)	-0.289 (0.190)	-0.004 (0.008)	0.008** (0.004)	-0.016 (0.041)	-0.007** (0.003)
N	1,104,252	1,104,252	1,104,252	1,104,252	1,104,252	1,104,252
Outcome mean	0.52	49.58	0.62	0.24	4.71	0.97
R^2	0.29	0.13	0.18	0.20	0.89	0.78
	Percent White	Percent Black	Percent Hispanic	Percent with No HS Degree	Percent that Walk to Work	Cars per Household
Distance to polling place (mi)	-0.004 (0.004)	0.001 (0.004)	-0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	-0.002 (0.007)
N	1,104,252	1,104,252	1,104,252	1,104,252	1,104,252	1,104,252
Outcome mean	0.65	0.25	0.09	0.15	0.06	1.34
R^2	0.91	0.94	0.85	0.86	0.85	0.92
B. Georgia						
	Female	Age	Democrat	Republican	Median HH Income	Urban
Distance to polling place (mi)	-0.0025*** (0.0009)	-0.1394 (0.1382)	-0.0020 (0.0020)	-0.0010 (0.0016)	0.0189 (0.0165)	-0.0120*** (0.0023)
N	548,392	548,392	548,392	548,392	548,392	548,392
Outcome mean	0.52	47.16	0.09	0.07	5.45	0.86
R^2	0.33	0.11	0.09	0.14	0.86	0.86
	Percent White	Percent Black	Percent Hispanic	Percent with No HS Degree	Percent that Walk to Work	Cars per Household
Distance to polling place (mi)	-0.0005 (0.0030)	0.0012 (0.0029)	-0.0008 (0.0010)	-0.0010 (0.0008)	-0.0004 (0.0003)	0.0107*** (0.0028)
N	548,392	548,392	548,392	548,392	548,392	548,392
Outcome mean	0.55	0.36	0.07	0.14	0.02	1.75
R^2	0.77	0.80	0.59	0.82	0.71	0.85

Note: This table reports coefficients for the travel route distance to polling place (mi). The outcome variables are the individual-level and census block-level covariates listed in each column. Each regression includes fixed effects for matched pairs of voters. Standard errors allow for clustering at precinct and segment level.